MoCHII Among the Monte Carlo Photoionization Codes: An Algorithmic and Physical Comparison with M^3, MAPPINGS V 5.2.1, IONIZE, CMACIONIZE, TORUS, and MOCASSIN

Abstract

This memo compares MOCHII (MOnte Carlo for H II regions) with the Monte Carlo photoionization codes closest to it in method, and with the deterministic one-dimensional plasma code two of them descend from. The principal pairwise comparison is with M^3 (Messenger Monte Carlo MAPPINGS V) and its parent MAPPINGS V 5.2.1, whose local trees are `~/RT_Codes/MAPPINGS/M3-main` and `~/RT_Codes/MAPPINGS/mappings_V-521`. Section 5. adds three further three-dimensional codes: IONIZE, the code of Wood, Mathis & Ercolano (2004) that supplies the WME column of the Lexington benchmark tables; CMACIONIZE, which its own source describes as the C++ version of that code; and TORUS, a radiation-hydrodynamics code whose photoionization loop borrows from both IONIZE and MOCASSIN. MOCASSIN is MOCHII's quantitative validation reference and appears throughout. Section 10.2 compares the one emission process all of them compute, the hydrogen recombination lines, together with CLOUDY c25.00, which is the one code in reach that solves the hydrogen atom instead of reading a table. There are four method classes among the seven, and no two codes agree on how the case A/B question should even be posed.

The comparison rests on connected execution paths and quantitative tests rather than on routine names. MOCHII is strongest as a modern three-dimensional transport and synthetic-observation engine: octree AMR, an analytic absorption estimator, a dedicated Cartesian DDA backend, solution-driven ionization-front refinement, dust absorption and scattering, stochastic dust/PAH emission, and observer peel-off imaging; and it is the only code here whose thermal balance and emergent line list come from the same level populations at the same electron density. MAPPINGS V 5.2.1 is strongest as a deterministic one-dimensional general plasma engine: a 30-element option, density-dependent multilevel cooling, finite-source-age photoionization, non-equilibrium cooling, radiation pressure, and steady radiative MHD shocks with an iterated precursor. It also has the most physical hydrogen-line treatment of the six three-dimensional and one-dimensional nebular codes compared here, a case parameter computed from the local Lyman-line escape probability rather than chosen. M^3 occupies a distinct middle ground: it grafts three-dimensional Monte Carlo transport onto an older MAPPINGS V 5.1.13-era physics tree, so it combines broad MAPPINGS microphysics with arbitrary Cartesian geometry, but it does not inherit all later 5.2.1 changes or expose all parent modes through its public 3D driver.

The limitations are correspondingly different. MOCHII has an H II-region-centered ion inventory; M^3 has no AMR, a communication-heavy uniform-grid packet algorithm, and a 3D driver that hardcodes pure case B; MAPPINGS V 5.2.1 is restricted to spherical or plane-parallel 1D structures and uses an outward-only diffuse field; IONIZE is serial, dust-free, and has no convergence test; CMACIONIZE has no global convergence criterion, no dust in the photoionization path, and a temperature solve that is off by default; and TORUS has the strongest transport and parallelism of the family alongside the least complete gas microphysics, missing He II photoheating and emitting no helium line at all. The recommended strategy is to retain the MOCHII transport architecture, use MAPPINGS V 5.2.1 as the current microphysics and 1D shock/time-dependence reference, use M^3 to study how MAPPINGS physics behaves after being mapped into three dimensions, and read TORUS for transport and parallel design rather than for gas physics.

Contents

1. Purpose, scope, and basis of the comparison

MOCHII and M^3 are self-consistent three-dimensional Monte Carlo photoionization codes, whereas MAPPINGS V 5.2.1 is a deterministic one-dimensional plasma and radiative-shock code. MOCHII adds gas physics to the validated MoCafe v2.00 dust radiative-transfer engine. Its founding design treats grid traversal, radiation estimators, dust physics, parallelism, file formats, and synthetic observations as first-class components. M^3 adds three-dimensional Monte Carlo radiation transport to MAPPINGS V, whose principal strength is its extensive atomic, ionic, heating, cooling, and emission-line physics [1, 3]. MAPPINGS V 5.2.1 continues that parent line as version 5.2.1, with separate photoionization, shock, equilibrium-cooling, non-equilibrium-cooling, and single-ion diagnostic drivers.

This memo distinguishes three levels of evidence:

1. *Physics present in the parent code.* The MAPPINGS V source tree contains one-dimensional photoionization, shock, non-equilibrium, dust, and plasma utilities.
2. *Physics connected to the three-dimensional driver.* A routine appearing in the source tree is not automatically active in the `montph7` Monte Carlo path.
3. *Physics quantitatively gated in the current code.* This is the strongest evidence: an analytic test, independent solver, database comparison, or published benchmark with a stated numerical tolerance.

This distinction is essential for shocks and time dependence. The public MAPPINGS tree contains `shock2--shock5`, `timion`, and `timtqui`; however, the three-dimensional `montph7` family sets `nmod='EQUI'` before the cell thermal solve. Thus the present three-dimensional M^3 calculation is an equilibrium photoionization model. The legacy routines remain valuable MAPPINGS capabilities, but they should not be described as coupled three-dimensional M^3 physics without additional development and validation.

No direct same-input, same-atomic-data timing or line-flux campaign among the three local executables has yet been performed. Numerical performance comparisons in this memo therefore use algorithm structure and each code's own recorded gates. They are diagnostic rather than a substitute for a controlled cross-code experiment.

2. Lineage: why MAPPINGS V 5.2.1 is not M^3 5.2.1

The three code names can easily obscure the most important architectural fact. M^3 is not a three-dimensional front end to the newly added MAPPINGS V 5.2.1 directory. Its source identifies itself as MAPPINGS V 5.1.13 with a separate Monte Carlo development history. The two trees still share 78 Fortran filenames, but the shared implementations have diverged substantially; the 5.2.1 versions of core files such as `photo6.f`, `photo7.f`, `shock5.f`, `mapinit.f`, and `cool.f` contain extensive later changes. M^3 adds its own `montph7`, Cartesian packet transport, PDF, and three-dimensional thermal-solver files instead [1, 4].

This gives each branch a different meaning:

- MAPPINGS V 5.2.1 is the newer parent-physics branch. Its public `map52` executable offers P6/P7 multizone photoionization, S5 shock plus precursor models, CIE and NEQ cooling curves, PIE curves, single slabs, and multilevel-ion diagnostic modes.

- M^3 is the three-dimensional geometry branch. It is the relevant comparison for arbitrary density fields, but its physics inventory must be assessed from the connected `montph7` path, not inferred from the presence of a similarly named MAPPINGS routine.
- MOCHII is an independent three-dimensional transport branch built from MoCafe rather than MAPPINGS. Similar physical terms therefore have independent algorithms, atomic-data pipelines, and validation histories.

Consequently, a discrepancy between MOCHII and M^3 cannot safely be labeled a discrepancy with MAPPINGS V 5.2.1 5.2.1. For future differential tests, 5.2.1 should be the primary one-zone and 1D microphysics reference; M^3 should be treated as the older MAPPINGS-derived 3D realization.

3. MAPPINGS V 5.2.1 in its own right

3.1 Deterministic 1D integration and diffuse radiation

P6 and P7 integrate successive zones in spherical or plane-parallel geometry. Step sizes respond to optical depth and predicted changes in temperature and ionization. The result has no Monte Carlo shot noise and is therefore efficient for large parameter grids, detailed spectra, and thin cooling zones. Isobaric models can include radiation pressure, $P(r) = P_0 + P_{\rm rad}(r)$, while other choices provide constant density or a prescribed density law. The model may terminate at a distance, an ion or hydrogen column, or an optical/radiation-bounded condition.

The price is geometric closure. The diffuse field is explicitly labeled "OUTWARD ONLY" in `photo6.f` and `photo7.f`. It follows the allowed 1D directions rather than transporting recombination photons over arbitrary solid angle. This is fast and often adequate for a monotonic sphere or slab, but it cannot represent porous escape, oblique shadows, clump illumination, or inward/backward diffuse coupling as a true 3D transport calculation can. The word "adaptive" in the S5 menu refers to an adaptive one-dimensional shock integration, not to a 3D AMR mesh.

3.2 Time dependence, cooling flows, and shocks

Unlike the public M^3 3D driver, the P6/P7 path genuinely connects a finite source lifetime to the time-dependent ion solver. It supports an equilibrium solution, a finite-age ionization calculation, and a post-equilibrium source-off evolution. The implementation advances ionic populations with adaptive substeps and couples the elapsed ionization-front time to each new zone. This is a useful one-dimensional ionization-history model, although it is not radiation hydrodynamics and does not evolve an arbitrary multidimensional density field.

The S5 path is an even clearer 5.2.1 advantage. It solves a steady, plane-parallel radiative MHD shock, adapts its step to atomic, cooling, and photon-absorption timescales, and iterates the shock radiation with an automatically calculated photoionized precursor. The source description advertises shock speeds from roughly 10 to 10^4 km s^{-1}. Companion drivers compute CIE cooling, non-equilibrium cooling, and PIE curves. For one-dimensional shock spectra and cooling flows, neither current 3D code offers an equivalent connected and mature path.

3.3 Atomic, electron, and dust physics

The standard preference file activates 16 elements, while `mapFull.prefs` activates all 30 elements from H through Zn; static dimensions permit 31 ion stages and 10,240 photon-energy entries. The thermal loop directly calls fine-structure, intercombination, resonance, multilevel, iron, recombination, free-free, collisional-ionization, charge-exchange, grain/PAH, and cosmic-ray terms. Optional κ electron distributions modify excitation, ionization, and recombination rates. These are major strengths for broad plasma diagnostics and for gas near or above coolant critical densities.

The 1D dust path is materially more complete than the public M^3 3D dust branch. In addition to grain charge, depletion, photoelectric heating, gas-grain collisions, PAHs, and destruction prescriptions, P6/P7 call `dusttemp` zone by zone. That routine implements a Draine–Li stochastic temperature distribution and returns an infrared photon field. Thus MAPPINGS V 5.2.1 has a coupled 1D gas–dust–IR calculation, whereas MOCHII has the stronger spatially resolved 3D dust transport and imaging chain.

Several cautions remain. The atomic tables are broad but globally coupled and are harder to provenance or replace ion by ion than the MOCHII registry. A Compton routine exists but no call from the examined main cooling path was found, and the microturbulent heating block in `cool.f` is commented out; neither should be counted as active 5.2.1 physics without a dedicated execution test. Resonance-line attenuation uses escape/transfer approximations rather than a multidimensional, frequency-redistributing Monte Carlo line solver.

3.4 Software and workflow limitations

MAPPINGS V 5.2.1 preserves a productive but legacy Fortran architecture: common blocks, compile-time maxima, runtime table loading, interactive menus, and many ASCII output products. P7 exposes experimental routines and an API, but the normal workflow remains less scriptable and less modular than MOCHII's namelist, HDF5/FITS, module, and Python-reader design. The documentation overview describes the manual as work in progress, so reliable use still requires source reading. The benefit of this conservatism is continuity with decades of MAPPINGS models; the cost is a higher barrier to automated regression testing, embedding, and isolated replacement of atomic data.

4. Executive comparison

Table 1: High-level comparison of the current three-dimensional codes.

Topic	MOCHII	M^3
Grid	Octree AMR or a memory-minimal uniform Cartesian backend; optional solution-driven I-front refinement.	Uniform Cartesian grid divided into equal MPI blocks; no AMR.

Table 1 continued.

Topic	MoCHII	M^3
Absorption transport	Analytic leaf-by-leaf attenuation estimator when the band has no scattering; only direction and frequency are sampled.	Lucy-style random optical depth, discrete absorption, and immediate local re-emission.
Diffuse field	Explicit selected H/He recombination channels rebuilt from the current state; case B OTS remains available.	Local continuum and line emissivity PDF sampled immediately after gas absorption; isotropic re-emission.
Frequency treatment	Dynamically allocated FUV and EUV segments, normally tens of rate-optimized bins with threshold alignment.	Up to 10,240 bins; a fixed number of packets is launched at every frequency and packet energy follows the source spectrum.
Parallel model	Packet decomposition and shared-memory grid arrays; one full grid copy per node.	Spatial domain decomposition; packets crossing blocks are synchronized through MPI.
Gas microphysics	Modern, provenance-bearing fitted data for an H II-region/PDR-centered registry; low-density cooling fits in the thermal loop.	Broad MAPPINGS ion network and multilevel line cooling inside the thermal loop; older but extensively exercised atomic-data compilation.
Dust	EUV/FUV absorption and HG scattering, live dust/PAH survival, equilibrium temperature, SEDust stochastic emission, and IR maps.	Rich grain charge, depletion, photoelectric, and PAH microphysics; the public 3D dust transport branch treats dust extinction/absorption but does not expose an equivalent scattering plus stochastic-IR imaging chain.
Synthetic observations	Leaf emissivities, flux-conserving line and field maps, dust-band maps, multi-observer peel-off images, and spectral cubes.	Extensive line spectrum and cell-resolved emissivities; maps can be made by projection, but no equivalent general observer peel-off layer is evident in the public path.
Best current use	Dusty, clumpy, spatially resolved H II regions and atomic PDR skins on simulation-derived density fields.	Broad nebular line diagnostics, denser gas, harder PN-like spectra, and problems that benefit from the full MAPPINGS ion and cooling inventory.

The concise verdict is:

MOCHII is presently the stronger three-dimensional transport, dust, AMR, and synthetic-observation framework. M^3 is presently the stronger broad-spectrum nebular microphysics framework, especially when many ion stages or density-dependent line cooling determine the solution.

Table 2: What MAPPINGS V 5.2.1 adds to the original two-code comparison.

Topic	MAPPINGS V 5.2.1 strength	Limitation relative to the 3D codes
Photoionization	Noise-free adaptive 1D integration; equilibrium, finite-age, and source-off evolution; radiation-pressure options.	Only spherical or plane-parallel structure; outward-only diffuse field.
Shocks and cooling flows	Connected steady radiative MHD shock plus iterated precursor, CIE cooling, NEQ cooling, and PIE modes.	Steady or prescribed 1D histories, not multidimensional radiation hydrodynamics.
Microphysics	Full 30-element preference set, many ion stages, density-dependent line cooling, κ electrons, cosmic rays, and broad spectral output.	Global tables are harder to audit and update ion by ion; not every dormant routine is connected to the main cooling path.
Dust	Grain/PAH charge and destruction, gas–grain exchange, stochastic temperature distribution, and a coupled 1D IR field.	No arbitrary 3D dust geometry, shadowing, AMR, or observer image construction.
Numerics and workflow	Fast deterministic parameter grids with decades of model heritage.	Legacy interactive common-block architecture, compile-time maxima, fragmented ASCII products, and incomplete manuals.
Best current use	Integrated spectra of 1D H II regions, PNe, AGN slabs, radiative shocks, and equilibrium/non-equilibrium cooling sequences.	Not a replacement for 3D modeling when geometry is part of the physical question.

5. The Monte Carlo photoionization codes closest in method

M^3 is not the only three-dimensional Monte Carlo photoionization code this comparison has to account for. Three others are close enough in method that MOCHII's choices are best read against them: IONIZE, the code of Wood, Mathis & Ercolano (2004) that supplies the WME column of the Lexington benchmark tables; CMACIONIZE, which its own source calls "the C++ version of Kenny Wood's photoionization code" (`src/CMacIonize.cpp`:392–393); and

TORUS, a radiation-hydrodynamics code whose photoionization loop borrows from both IONIZE and MOCASSIN. MOCASSIN itself is MOCHII's quantitative validation reference and appears throughout this memo rather than as a separate subsection.

The evidence standard of Section 1. applies unchanged: every statement below was read in the source, and where a capability exists in a tree but is not reached from the photoionization driver it is labeled as such. The *cooling* inventories of these same four codes were audited separately and are not repeated here; that audit, with its file and line references and the defects it found on each side, is `docs/MoCHII_cooling_analysis.pdf`. This section covers the other axes.

5.1 IONIZE: the reference implementation of the method

IONIZE is the ancestor of this family and the most instructive comparison precisely because it is small: 33 fixed-form Fortran files, one executable, no libraries.

Packets carry photon number rather than energy, and the band is 13.6–54.4 eV only (`planckset.f`:19–21). The estimator is Lucy's path-length form, but *frequency-unresolved*: as a packet crosses a cell, the code accumulates the rate integrals themselves — one accumulator for each species it knows about, `jmeanH`, `jmeanHe`, `jmeanN(...,1:3)` and so on (`tauint.f`:184–268) — and never stores a mean intensity. There is no $J_\nu(\nu, \text{cell})$ array anywhere in the code. The consequence is structural rather than numerical: a rate integral for any species not in that hardcoded list cannot be recovered after the run, so adding an ion is a code operation. MOCHII's `jt_ion(nnu,nleaf)` band tally is the opposite choice, and it is what makes its registry a data operation.

The opacity is hydrogen and helium only (`tauint.f`:159–161); the metals are tallied but never attenuate. The diffuse field, by contrast, is fully explicit and physically consistent: the balance uses the Verner & Ferland case-A total recombination coefficient (`recomb.f`:13–15, 4.19×10^{-13} cm^3 s^{-1} at 10^4 K), and after each absorption the packet is re-emitted as a ground-state continuum photon with probability α_1/α_A, sampled from a Saha–Milne emissivity distribution, with four helium channels and He I Lyman α handled on the spot (`probset.f`:28–60, `mcionize.f`:278–339).

The grid is a uniform Cartesian mesh with 65^3 compiled in (`grid.txt`:3), and the density is a hardcoded analytic function of position evaluated on cell centers (`density.f`:15–20); there is no grid-file reader, so a simulation-derived density field cannot be read at all. The code is *serial*: no MPI and no OpenMP appear in any source file. Seven elements and 21 stages are followed, without He^{++}, and with C^0 and S^0 assumed absent; n_e comes from ionized hydrogen and helium only, which the source states outright (`tbal1.f`:39–40).

Two properties of its iteration are worth recording because MOCHII spent real effort on both. There is *no convergence test* — the manual prescribes running many iterations and comparing successive ones by eye — and the ionization is iterated alone for the first four iterations, the thermal balance starting only at the fifth (`itbal.f`:27–41). The temperature solve is a logarithmic secant on the gain/loss ratio (`tbal1.f`:78–107); below 4000 K the cell is declared neutral and pinned at 500 K.

There is no dust. The only grain-related term is a scalar gas-heating factor $3 \times 10^{-25}\, n n_\mathrm{e}$ scaled by an input parameter that the shipped deck sets to zero (`ioneng.f`:56–68). Outputs are a direction-binned escaping spectrum on a 20×20 grid of (θ, ϕ) and an emissivity cube; there is no imaging, and the manual's recipe for a map is to sum the cube along an axis. The sibling directory `diffuse/` in the same distribution is a different code, a dust-scattering polarization Monte Carlo with Stokes peel-off imaging and no gas physics; it shares only utility routines and should not be read as IONIZE's imaging stage.

5.2 CMACIONIZE: the same method, modernized

CMACIONIZE inherits IONIZE's physics and rewrites its software. The transport is a random optical depth with the Lucy path-length estimator (`src/IonizationPhotonShootJob.hpp`:135, `src/DensityGrid.hpp`:150–197), the diffuse re-emission is the Wood–Mathis–Ercolano scheme reimplemented term by term (`src/PhysicalDiffuseReemission-Handler.cpp`:52–203), and the hydrogen line coefficients are numerically identical to IONIZE's (Section 10.2). What changed is everything around the physics.

Frequencies are continuous: a packet's cross sections are evaluated once at emission (`src/PhotonSource.cpp`:230–244) and there is no band grid in transport at all. This removes the discretization question that Section 6.3 shows to be the largest single lever on MOCHII's helium and metal ionization, and replaces it with sampling noise in the rate integrals. Three grid types are offered — uniform Cartesian (the default), an octree AMR with Morton-style keys, and a Voronoi mesh that can move with the fluid — plus a fourth structure, a subgrid decomposition used by the task-based mode. Refinement can be static or solution-driven, on either cell opacity or O I content (`src/OpacityAMRRefinementScheme.hpp`:90–99, `src/OIAMRRefinementScheme.hpp`:87–94). Two gaps are visible in the source: `AMRRefinementScheme::coarsen` is declared and has no caller anywhere, so the mesh can only ever get finer, and the MPI branch of the refinement carries the comment `/// WE HAVE TO DO SPECIAL THINGS FOR MPI HERE!` (`src/AMRDensityGrid.hpp`:337).

Parallelism is OpenMP through a job market plus MPI as a photon-count decomposition over a *fully replicated* grid, with the accumulators reduced across ranks and the cell solve split by contiguous cell block (`src/IonizationSimulation.cpp`:395–397, 458–617). That is the same division of labor as MOCHII's, which also replicates the grid and distributes packets; TORUS is the one code here that decomposes the grid itself. MPI is refused at startup for the dust, hydrodynamic, emissivity and task-based modes.

Three properties bear directly on the comparisons this memo makes elsewhere. The metal line emissivities are solved from level populations at the local electron density (`src/LineCoolingData.cpp`:1785–1798), which is the axis MOCHII closed with `par%cooling_model = 'local_ne'`. The temperature solve is *off by default* and, when on, is capped at 30 000 K for a stated reason — "helium charge transfer rates are only valid until 30,000 K" (`src/TemperatureCalculator.cpp`:831–833). And there is *no global convergence criterion of any kind*: the iteration is `while (loop < _number_of_iterations)` (`src/IonizationSimulation.cpp`:359), with a plain input count, no cell-maximum test, no integrated measure and no early exit. Metals carry neither opacity nor free electrons, there is no collisional ionization anywhere in the tree, and He^{++} does not exist as a species.

Dust is absent from the photoionization path — a case-insensitive search of the ionization, temperature and state-calculator sources returns nothing. It exists instead as a separate hardwired mode whose own documentation calls it a first-year lab project (`src/CMacIonize.cpp`:50–53), with grey per-band constants, a forced first interaction, single-observer peel-off and Stokes imaging. There is no dust temperature and no infrared emission anywhere in the tree.

Synthetic observations are the weakest axis. The ionization mode offers point "trackers" that record the absorption or the spectrum inside a named cell, and grid dumps whose field list contains no emission-line entries. A line-of-sight emission integrator, `get_total_emission`, is implemented for all three grid types and *has no caller*; the emissivities it would integrate are themselves produced only by a separate post-processing mode, which is Cartesian-only and rejects MPI. Two statements in the tree's own documentation are stale and should not be quoted from it: the claim that the code is parallelized with OpenMP only (`src/CMacIonize.cpp`:416–420), contradicted by the working MPI code, and a comment implying a Python library replaced the in-run emissivity calculation when the compiled mode is the fuller route.

5.3 TORUS: photoionization inside a radiation-hydrodynamics code

TORUS is the most capable of the three in transport and the least specialized to photoionization, which shows in both directions.

The estimator is again Lucy's, with the path length weighted by $\ell w/(h\nu)$ and accumulated for each ion (`photoionAMR_mod.F90`:6679–6692); a modified random walk handles the optically thick regime. Dust scattering uses real Mie phase matrices rather than a Henyey–Greenstein approximation (`:3283`). The diffuse field is transported by default, and the switch that turns it off removes only the ionizing continuum from the redraw spectrum (`:3347--3349`), leaving the sub-Lyman lines and the dust continuum in place — so "on the spot" means something narrower here than in MOCHII. Independently of that, a `caseB` flag replaces the hydrogen recombination coefficient with the *constant* 2.7×10^{-13} cm^3 s^{-1}, with no temperature dependence (`photoion_utils_mod.F90`:305–310); setup only warns if it is combined with a transported diffuse field. This is exactly the conflation MOCHII separated into `par%case_ab` and `par%line_case`, left here as a warning rather than as two parameters.

The frequency grid is 1000 logarithmic bins from 1 mm to about 124 eV with four extra points pinned at $\pm 1\%$ around the helium edges (`:2124--2151`) — threshold *bracketing*, the same concern as MOCHII's threshold alignment, addressed by refining around the edge rather than by placing a bin boundary on it.

The grid is an AMR tree that can be one-, two- or three-dimensional (binary, quad, octree), with a cylindrical option and a two-dimensional Cartesian slab whose missing third dimension is compensated by re-weighting the emission direction. Solution-driven refinement is live and physically the same criterion MOCHII uses: a relative neutral-hydrogen-fraction jump across each face against a tolerance (`hydrodynamics_mod.F90`:14582–14618). One switch is dead and should not be cited as a capability: `doPhotoRefine` is read from the input file and its only use site is `if (doPhotoRefine .and. 0 == 1)` (`:4428`).

Parallelism is the most developed of the four: MPI decomposes the grid itself, with packets shipped between ranks in batched stacks, plus extra ranks assigned as duplicates of busy domains apportioned by measured packet crossings (`loadbalance_mod.F90`:16–219), and OpenMP threads the per-cell equilibrium solve (`:3855--3864`). MOCHII distributes only transport and keeps one grid copy per node; TORUS distributes the state.

Convergence is a nested pair: an inner alternation of ionization and thermal solves until both change by less than 1% in every subcell, and an outer test on the *cell-maximum* fractional temperature change below 0.05 (`:4324--4339`). There is no volume-integrated measure. What TORUS does instead, when the outer test stalls, is diagnose the cause: cells the Monte Carlo has undersampled are counted, and finding any of them doubles the packet count and continues (`:4344--4347`). That is an answer to the same noise-floor problem MOCHII solved with `par%conv_crit = 'vol'`, from the other end — raise the photon budget until the cell-maximum test becomes meaningful, rather than change the measure.

The heating inventory is exactly three terms: H I photoionization, He I photoionization, and dust (`:7810--7823`). He II photoheating is absent although He^{++} is tracked, and there is no metal photoheating and no grain photoelectric heating in this balance. Recombination data are Verner & Ferland for H and He and Pequignot et al. (1991) plus Nussbaumer & Storey (1983) for the metals — *the* MOCASSIN *set*, so the in-code swap experiment reported in Section 8.2 transfers to TORUS unchanged. Its H$^+$ and He^{++} coefficients agree with MOCHII's default α_A to four digits; only He$^+ + e$ differs, and there TORUS sits 4.7% *above* MOCHII, the opposite side from MOCASSIN's low value. One provenance curiosity: the free-bound γ tables TORUS reads are byte-identical to MOCASSIN's and MOCHII's copies, but its own comment attributes them to Ferland (1980) where the other two carry the Ercolano & Storey compact format. The numbers are not in question, only the citation.

Dust is coupled to the gas in both directions, which no other code in this comparison does: grain emission enters the gas cooling side and grain heating the gas heating side of the same balance, or, decoupled, a gas–grain collisional exchange term carries the energy between them (`:6604--6628`, `:6253--6271`). The infrared emission is a single equilibrium temperature per cell; there is no stochastic single-photon heating and no PAH channel on this path, so MOCHII's SEDust coupling and its $x_{\rm HII}$-weighted PAH share have no counterpart.

Imaging is genuine peel-off with the full Stokes vector and $e^{-\tau}$ to the boundary, cloned at emission and after every scattering (`:9109--9113`, `image_mod.F90`:187–188), for one of four image types (free-free, forbidden line, recombination line, dust). It is monochromatic: the photoionization image path carries a fixed wavelength and applies no Doppler shift. TORUS does have a velocity-resolved image adder, but it belongs to the atomic and molecular line transfer modes and is not wired to the photoionization loop — present in the tree, not connected. There is no emergent SED and no datacube from this loop.

5.4 What the four have in common, and where MOCHII sits

Table 3: The Monte Carlo photoionization codes closest to MOCHII in method, on the axes not covered by the cooling audit. Entries are what the photoionization path does, not what exists in the tree.

Topic	IONIZE	CMACIONIZE	TORUS	MOCHII
Estimator	Lucy path length, rate integrals accumulated directly; no J_ν stored	Lucy path length, continuous frequency	Lucy path length, 1000-bin grid	Analytic edge walk without scattering, band-resolved J_ν tally
Grid	Uniform Cartesian, size compiled in; analytic density only	Cartesian / octree AMR / Voronoi / subgrid	AMR in 1D, 2D or 3D, cylindrical option	Octree AMR or raster Cartesian with a DDA walk
Solution-driven refinement	none	opacity or O I criterion; no coarsening	$x_{\rm HI}$ gradient across faces	$x_{\rm HI}$ front flags plus face-neighbor jump
Parallel model	serial	OpenMP + MPI over packets, grid replicated	MPI over the grid with load-balancing ranks, OpenMP over cells	MPI over packets, shared-memory grid per node

Table 3 continued.

Topic	IONIZE	CMACIONIZE	TORUS	MOCHII
Diffuse field	explicit, case A, four He channels	the same scheme, opt-in, default off	transported by default; `caseB` is a constant coefficient	explicit H/He channels plus the He I excited channel; case-B OTS available
Global convergence	none (compare iterations by eye)	none (fixed iteration count)	cell-maximum ΔT_e, packet doubling on undersampling	cell-maximum or volume-integrated, tolerance-gated
Metals	tallied, no opacity, no electrons	tallied, no opacity, no electrons; no collisional ionization	22 ions with charge exchange, no collisional ionization	registry with opacity, optional electrons and heating, two collisional-ionization models
Dust	none	none on this path	absorption, Mie scattering, equilibrium $T_{\rm dust}$, gas–grain exchange	absorption, scattering, live survival, $T_{\rm dust}$, stochastic PAH emission
Images	escaping spectrum by direction bin; emissivity cube	trackers only; the line-of-sight integrator is never called	Stokes peel-off, monochromatic, four image types	multi-observer peel-off, direct/scattered split, bin-resolved cubes, flux-conserving line maps

Read across that table, the family divides on one line. IONIZE and CMACIONIZE are photoionization codes in which the radiation field is a means to the ionization state, and their synthetic-observation layer is correspondingly thin. TORUS is a transport code with photoionization added, which is why its imaging and parallelism are the strongest here and its gas microphysics the most incomplete — missing He II photoheating and, as Section 10.2 shows, missing helium lines entirely. MOCHII sits with TORUS on transport and observation and with MOCASSIN on gas microphysics, and it is the only one of the five whose thermal balance and emergent line list are computed from the same level populations at the same electron density.

6. Radiative-transfer algorithms

6.1 Radiation-field estimators

Both three-dimensional codes estimate the cell radiation field from packet trajectories, but their estimators differ fundamentally. M^3 follows the Lucy path length estimator. At frequency ν, a packet with energy ϵ_ν crossing a distance l

in cell volume V contributes schematically

$$J_\nu = \frac{1}{4\pi\,\Delta t\,V}\sum_{\text{paths}}\epsilon_\nu l. \tag{1}$$

The distance to the next absorption is sampled from

$$\tau_p = -\ln p, \tag{2}$$

with p uniform on $(0,1)$. The packet crosses Cartesian cells until its accumulated optical depth reaches τ_p, is absorbed, and is re-emitted in situ from the local emissivity distribution. This is conceptually general and treats diffuse radiation naturally.

For the absorption-only ionizing band, MOCHII instead integrates the attenuated packet weight analytically through every crossed leaf:

$$\Delta J_b(\text{leaf}) = wL_{\text{pkt}}\frac{e^{-\tau_{\text{in}}}-e^{-\tau_{\text{out}}}}{\kappa_b}. \tag{3}$$

The packet is followed to the domain edge and contributes the expectation value of the absorbed path in each leaf. There is no sampled absorption location; statistical noise comes only from direction and bin sampling. The analytic attenuation gate measured a mean photo-rate bias of +0.004%. An optional Owen-scrambled Sobol launch (`launch_sequence='sobol'`) stratifies exactly those remaining launch variables: on the same gate the cell-wise photo-rate error then scales as $\sim 1/N$ rather than $1/\sqrt{N}$, with measured effective photon gains of 29$\times$ at 2^{17} packets and 222$\times$ at 2^{20}, and the launch set becomes independent of the MPI task count. When dust scattering is enabled, MOCHII changes to a forced-first-interaction estimator, applies the dust albedo as a packet weight, samples an HG scattering direction, and uses Russian roulette at low weight.

Equation (3) is a substantial advantage for the present MOCHII target regime: ionizing radiation in H II regions is normally absorption-dominated. M^3 pays the variance of discrete absorption events even in that limit. Conversely, the M^3 packet cycle is more general for a diffuse spectrum containing many continuum and line channels, whereas the current MOCHII diffuse emitter is explicitly limited to selected recombination channels.

MAPPINGS V 5.2.1 has neither Monte Carlo estimator. Its zone integrator is deterministic and therefore essentially noise-free, but achieves that efficiency by imposing spherical or plane-parallel symmetry and an outward-only closure for the diffuse field.

6.2 Diffuse radiation and energy packets

In M^3, gas absorption converts the stellar packet into a diffuse packet immediately. Its new frequency is drawn from a local emissivity PDF containing recombination continua and lines, and its new direction is isotropic. This implements a full three-dimensional alternative to outward-only diffuse-field approximations [1].

MOCHII offers two controlled limits. Case B uses on-the-spot recombination. With the explicit diffuse field enabled, case A is used and the emission table is rebuilt each iteration from the current state. The principal channels are H II, He II, and He III ground recombination continua; the optional He I excited channel adds the 19.82-eV, 584-Å, and two-photon contributions. These packets use the same transport estimator as stellar packets. This split makes case-A/case-B bracketing and energy accounting transparent, but it is not yet the full local emissivity inventory sampled by M^3.

6.3 Spectral discretization

The public M^3 compile-time ceiling is 10,240 frequency entries. The Monte Carlo loop launches N packets for every frequency interval; packet energy is proportional to $L_\nu \Delta\nu / N$. This stratified sampling guarantees representation of weak spectral intervals and is attractive for detailed continua. Its cost scales approximately with $N_\nu N$ rather than with one total packet budget, and its large $J_\nu(x, y, z)$ and opacity arrays are major memory consumers.

MOCHII samples a frequency bin from the luminosity CDF and assigns every packet the same band luminosity. Its bins are designed around rate integrals rather than spectral oversampling. The ionizing grid can align edges with H, He, and active metal thresholds, while an independent FUV segment preserves the EUV resolution. This is efficient for ionization and thermal balance, but a fine line-rich diffuse spectrum would require many more bins and packets than the normal 32–64-bin production setup.

6.4 Velocity fields and line transport

M^3 reads a three-dimensional velocity field and applies a projected Doppler shift to the packet frequency bin during continuum transport. MOCHII currently has no velocity-dependent band transport. This is a real M^3 advantage for bulk-flow effects on continuum opacity.

None of the three codes should yet be treated as a general resonance-line transfer solver. M^3 includes resonance emissivities but, in the published implementation, assigns resonance lines the same cross section as the continuum. This is not a frequency-redistributing resonant-scattering calculation. MOCHII explicitly treats its diagnostic lines as optically thin emissivities and leaves resonance transfer to a dedicated code such as LaRT. MAPPINGS V 5.2.1 applies geometry-dependent escape and attenuation approximations rather than multidimensional redistribution. For forbidden and recombination lines this is normally adequate; for Lyα, C IV, Mg II, or line profiles in moving media, neither current path is sufficient.

7. Grid algorithms, resolution, and parallelism

7.1 Octree AMR versus a uniform Cartesian mesh

The primary spatial difference is adaptive resolution. M^3 accepts an arbitrary density distribution on a Cartesian mesh, but every cell at a given global resolution exists. The published HII20, HII40, and PN150 tests used 33^3 and 55^3 grids. Most integrated lines agreed within 10%, but I-front and boundary tracers such as [O I], [O II], and [S II] could differ by more than 10% [1]. This is the expected weakness of a uniform grid: the thin zone that most needs resolution occupies a small part of the volume.

MOCHII can ingest a pre-existing octree or rebuild the octree after an initial solution. A leaf is refined when it lies in the I-front or sees a large face-neighbor ionization jump. In the recorded gate, refinement from a level-5 start to level 7 produced 464,248 leaves instead of the 2,097,152 cells of a complete level-7 grid; the effective ionized radius agreed with the native level-7 result by 0.008%. The refinement follows clump skins and distributed fronts in turbulent media, not merely the source position.

For problems that truly require a uniform mesh, MOCHII also has a raster-indexed Cartesian backend. It allocates no octree topology, neighbor table, center array, or leaf maps. The dedicated incremental Amanatides–Woo DDA precomputes $t_{\max}$ and t_{Δ} per ray and advances by comparisons and additions. In the recorded Strömgren test it was twice as fast as the octree traversal, while matching the solution to rounding. At 512^3, omitting the tree and geometry arrays saves approximately 10 GB before physical fields are counted.

7.2 MPI decomposition and scaling trade-offs

M^3 spatially decomposes the Cartesian domain into equal blocks. Each rank stores the local thermal and ionization state, and packets crossing a block boundary are transferred through collective MPI operations. Spatial decomposition is attractive when no node can hold the full grid. It also introduces frequent synchronization: the current transport implementation collectively reduces large packet-state arrays containing flags, positions, directions, frequencies, and absorption counts. At large N_ν, packet count, and MPI task count, communication can dominate.

MOCHII decomposes packets rather than space. The octree and leaf fields reside in MPI-3 shared memory, so ranks on one node share one copy; each node nevertheless holds a complete grid. Ranks transport independent packets and reduce the accumulated radiation tally at the end of a pass. This minimizes packet communication and is well matched to Monte Carlo transport, but the requirement of a full grid on each node ultimately limits the largest possible AMR problem. Thus the scaling comparison is not one-sided: MOCHII should normally scale more cleanly in packet number, whereas M^3 can distribute spatial memory more aggressively. MAPPINGS V 5.2.1 avoids the 3D memory problem entirely by advancing one 1D structure, which is a decisive throughput advantage only when its symmetry assumptions are physically defensible.

8. Ionization, thermal balance, and atomic data

8.1 Element and ion inventories

The local M^3 default atom file activates 16 elements: H, He, C, N, O, Ne, Na, Mg, Al, Si, S, Cl, Ar, Ca, Fe, and Ni. Static array dimensions allow up to 30 elements and 31 ion stages. The published description characterizes the parent MAPPINGS V system as containing 30 elements and roughly 80,000 cooling and recombination lines [1]. The distinction between the parent capacity and the default public data set should be retained when quoting these numbers. The new MAPPINGS V 5.2.1 audit resolves the parent-code side directly: `mapStd.prefs` is the 16-element setup and `mapFull.prefs` activates 30 elements from H through Zn. This full preference set, rather than the M^3 default, is the relevant breadth reference.

MOCHII currently treats H and He plus an eleven-element metal registry: C, N, O, Ne, S, Ar, Mg, Fe, Si, Cl, and Ca. The tracked stages are chosen for H II regions and atomic PDR skins, rather than extending to the highest charge state of every nucleus. New ions are added through provenance-bearing element, cooling-fit, and n-level files. This is much easier to audit and extend than the global MAPPINGS tables, but the present inventory is less suitable for hard PN, X-ray, or coronal-ion problems.

8.2 Photoionization, recombination, and charge exchange

MoCHII uses exact hydrogenic ground-state cross sections for H I and He II, and Verner et al. [6] fits for He I and metals. Hydrogen and helium recombination defaults to the Badnell (2023) radiative total minus the Mao–Kaastra (2016) ground-state coefficient (case B), with the Hui–Gnedin fits retained as an option; metal RR and DR are derived primarily from Badnell/CHIANTI v11.0.2 data. Charge exchange uses the modern first-stage rates of Huang et al. and the higher-stage Kingdon–Ferland compilation. Each generated file records its source, fit range, formula, and maximum fit error.

M^3 uses the established MAPPINGS compilation. The published code description identifies photoionization data from Seaton, Raymond, Osterbrock, and Gould, including Auger corrections, and separates hydrogenic from non-hydrogenic recombination. Much of this compilation is older than the MoCHII sources, but it spans more ions and has been exercised in many MAPPINGS applications. “Newer” is not automatically equivalent to “more accurate”: line predictions can differ by several to tens of percent among reputable collision-strength and A-value compilations, especially for iron. The decisive test is therefore ion-by-ion comparison with direct database evaluation and observation, not vintage alone.

8.3 Electron distributions and secondary processes

Both M^3 and MAPPINGS V 5.2.1 support either a Maxwellian or a κ electron energy distribution, with rate corrections supplied by KAPPA data. This is useful when exploring non-Maxwellian interpretations of nebular diagnostics. MoCHII assumes a Maxwellian distribution.

All three codes include secondary-ionization concepts. MoCHII optionally partitions the energy of photoelectrons above 40 eV among heat and secondary H/He ionizations following Shull–van Steenberg. The MAPPINGS ion-balance routines call a secondary-ionization module and also include cosmic-ray ionization/heating. The MAPPINGS-derived codes therefore have the broader general-plasma framework. A Compton routine is present in both source trees, but no call from either examined main cooling path was found; it should not be counted as active physics without a dedicated test.

8.4 The critical difference: density-dependent metal cooling

This is the most important gas-physics advantage of both MAPPINGS-derived codes over MoCHII. Their thermal solvers call fine-structure, intercombination, multilevel, and iron line modules during every evaluation of the net cooling function. Level populations and collisional de-excitation therefore affect the equilibrium temperature when n_e approaches or exceeds a transition’s critical density.

MoCHII deliberately separates two products:

- analytic Tier-1 cooling fits, evaluated cheaply inside the thermal bisection in the optically thin low-density limit; and
- Tier-2 n-level statistical-equilibrium data, evaluated on the converged state to produce density-dependent diagnostic emissivities.

The output line ratios therefore respond to density, but collisional quenching does not feed back into the hot thermal loop. At ordinary H II-region densities this is often acceptable. In compact H II regions, dense PN, shocked cooling zones, or any model with $n_e \gtrsim n_{crit}$ for a dominant coolant, M^3 is physically better founded. Moving the generic MOCHII n-level solver or a reduced density-suppression fit into the thermal loop is the highest-value gas-physics improvement suggested by this comparison.

8.5 Thermal processes and equilibrium solution

All three codes solve ionization and temperature under the current radiation field, iterating the spatial solution where required. MOCHII brackets the root of heating minus cooling in $\log T_e$ and re-solves the H/He and metal ionization at every trial temperature. Its current budget includes photoionization heating, H/He recombination and collisional losses, free-free emission with a tabulated Gaunt integral (H/He and, by default, the trace metals with their net ion charge), metal-line cooling, optional metal photoheating, secondary ionization, and the atomic-PDR grain and neutral collision terms.

The MAPPINGS thermal solver used by MAPPINGS V 5.2.1 and inherited by M^3 includes a broader traditional plasma budget: hydrogen and helium line cooling, fine/intercombination/resonance and multilevel metal cooling, free-free, collisional ionization, recombination, photoionization heating, charge-exchange energy exchange, cosmic rays, grain/PAH photoelectric terms, and grain collisional cooling. The microturbulent dissipation block in the examined 5.2.1 `cool.f` is commented out and is therefore not listed as active. This breadth and the direct multilevel coupling are major MAPPINGS strengths.

9. Dust, PAHs, and PDR physics

9.1 Where the MAPPINGS microphysics is stronger

The MAPPINGS grain microphysics is detailed. The local source tree carries graphite and silicate size distributions, depletion bookkeeping, grain charge balance, electron and ion sticking, grain photoelectric heating, collisional grain cooling, PAH charge states, and optional thermal or ionization-parameter-dependent grain destruction. For the local exchange of energy and charge between gas and a resolved grain population, this is more microscopic than the current MOCHII Bakes–Tielens heating prescription.

In MAPPINGS V 5.2.1, this layer is connected to the P6/P7 zone integration and to `dusttemp`, which computes a Draine–Li stochastic temperature distribution and an IR photon field. It is therefore a complete 1D dust emission path, not merely a library capability. The distinction below is specific to the public M^3 three-dimensional branch.

9.2 Where the three-dimensional M^3 dust path is limited

The public `montph7_dust` branch includes dust in the extinction and terminates a packet when the selected interaction is grain absorption. The dust data contain absorption, scattering, and asymmetry information, but the examined 3D branch computes an absorption fraction and does not sample a new dust-scattering direction. The dedicated `dusttemp` routine is called from the one-dimensional `photo6/7` family, not from the public 3D driver. Consequently, the rich MAPPINGS grain microphysics should not be conflated with a complete 3D dust radiative transfer and emergent-IR calculation. This does not imply that dust is absent from M^3: dust opacity, depletion, grain heating/cooling, and PAH processes affect the gas. It means that the public path does not presently expose a validated chain equivalent to MOCHII's dust scattering, absorbed-energy ledger, stochastic emission, and observer maps.

9.3 The MoCHII dust chain

MOCHII couples dust to the same FUV/EUV radiation field that sets the gas state:

1. Gas and dust compete for ionizing photons using bin-dependent absorption cross sections. The dusty Strömgren gate reproduces the one-dimensional radius to +0.69% and the classic radius reduction $R/R_0 = 0.735$.
2. With scattering enabled, the extinction includes the dust scattering coefficient, a forced first event is sampled, and the packet follows an HG direction with an albedo weight. The dusty-sphere result lies between the pure-absorption and extinction-as-absorption bounds.
3. The live dust model updates the dust abundance from the computed neutral fraction. A separate PAH survival fraction permits destruction inside ionized gas and produces a PAH-bright I-front ring.
4. The absorbed FUV/EUV power gives a leaf equilibrium mixture temperature and an energy-conserving IR spectrum.
5. SEDust supplies astrodust, DL07, or Zubko stochastic emission. The final normalization enforces $\int L_\lambda \, d\lambda = \sum_{\rm leaf} H_{\rm dust} V$; the smoke test closes at unity and resolves the usual PAH bands.
6. Leaf SEDs at selected wavelengths generate spatially resolved 8, 24, 70, and 160-μm maps.

For dusty three-dimensional H II regions, this end-to-end chain is the clearest overall physical advantage of MOCHII.

9.4 PDR scope

MOCHII extends the band below 13.6 eV so that low-threshold metals and dust see the FUV field beyond the H I-front. Optional switches include electrons from the metal cascade, metal photoheating, Bakes–Tielens grain photoelectric heating plus recombination cooling, and neutral-H impact cooling in [C II] 158 μm and [O I] 63 μm. This is an atomic PDR skin, not a complete PDR code. H_2/CO chemistry, molecular self-shielding, molecular cooling, line transfer, and gas–dust collisional coupling remain outside the model.

The MAPPINGS trees likewise contain more grain and general plasma physics than a minimal H II solver, but the present comparison did not identify a full molecular H_2/CO network in either the 5.2.1 photoionization path or the public M^3 3D driver. None of the three codes should be selected as a substitute for a dedicated molecular PDR code without additional chemistry.

10. Emission lines, continua, and synthetic observations

10.1 Line inventories and emissivity accuracy

The MAPPINGS lineage was designed to predict a very broad line spectrum. The M^3 cooling and output path includes resonance, intercombination, forbidden, fine-structure, hydrogenic, helium, heavy recombination, and extensive iron-line data. Its five-level and multilevel populations are already part of the thermal solution. This makes M^3 attractive for integrated diagnostic grids and hard nebulae for which a large fraction of the periodic table and many ion stages matter. MAPPINGS V 5.2.1 is the strongest version of that argument when 1D geometry is acceptable: the full preference set covers 30 elements, and the deterministic zone solver can afford a detailed spectrum without Monte Carlo sampling noise. M^3 trades some parent-code recency and 1D special modes for arbitrary Cartesian geometry.

MOCHII has a narrower but highly auditable diagnostic layer. Every registered ion with a Tier-2 file is solved by the same partial-pivot statistical-equilibrium code using fitted CHIANTI v11 effective collision strengths and explicit A-values. H I and He II recombination lines use Storey–Hummer case-A/B tables. The nebular continuum includes free-bound plus free-free and two-photon components. On an identical state, the Fortran line luminosities match the independent Python solver to at least four significant figures. The limitation is coverage, not the internal emissivity algebra.

10.2 Hydrogen recombination lines

Hydrogen recombination lines deserve their own comparison, for two reasons. They are the normalization of every published line list, so an error in Hβ propagates into every ratio quoted against it; and they are the one emission process every code here computes, which makes them the cleanest available cross-section through the design choices of the family. CLOUDY c25.00 is carried in this subsection as well, because it answers the question in a different category from the other six. The seven implementations fall into four method classes.

Table interpolation in (T_e, n_e). MOCHII reads the Storey & Hummer (1995) case-A and case-B tables (11 temperatures $\times$ 13 densities for case B, 10 $\times$ 9 for case A) and interpolates bilinearly in $(\log T_e, \log n_e)$, clamped at the grid edges, for seven lines (Hα, Hβ, Hγ, Hδ, Paα, Paβ, Brγ). MAPPINGS V 5.2.1 and M^3 read the same Storey–Hummer calculation in a richer layout: an absolute Hβ emissivity on 12 temperatures $\times$ 9 densities for *each* case, spline-interpolated in $\log T_e$, times 90 line ratios (15 lines $\times$ 6 series) on their own 10×9 grid (`HRECDAT.txt`, `HRECRATA.txt`, `HRECRATB.txt`; `hydro.f`:378–459). The density grid reaches $10^{10}\,\mathrm{cm}^{-3}$, against 10^{14} in MOCHII's case-B file and 10^{10} in its case-A file. MOCASSIN's *output* path is also a table interpolation, of `e1bx.d` — the same Storey–Hummer file MOCHII parses — by fourth-order Lagrange in $\log n_e$ and $\sqrt{T_e}$, up to $n_{\mathrm{up}} = 30$.

Fixed ratios times a fitted Hβ. MOCASSIN's *iteration* path reads `r1b0100.dat`, whose footer records what it is: `!Te=1.e4 Ne=1.e2 Case B`. The ratios carry no temperature or density axis, so Hα/Hβ is fixed at the tabulated 2.86 whatever the local state, and the only temperature dependence of any H line is the Hβ normalization. TORUS does the same thing with the same numbers: a ratio table stated in its own comment to be "for $T = 10^4$ K and $n = 10^2\,\mathrm{cm}^{-3}$" times an H$\beta$ power law (`photoionAMR_mod.F90`:8022–8043). The consequence is worth naming, because it is a physical one: in both codes *the Balmer decrement cannot respond to temperature at all*. Against Storey–Hummer at $n_e = 100\,\mathrm{cm}^{-3}$ the frozen 2.86 is −5.9% at 5000 K, right at 10^4 K, +4.1% at 2×10^4 K and +5.9% at 3×10^4 K.

Analytic power-law fits. IONIZE and CMACIONIZE carry *numerically identical* coefficients — CMACIONIZE is the C++ rewrite, and this is where that shows most plainly:

$$\frac{4\pi j(\mathrm{H}\beta)}{n_\mathrm{e} n_\mathrm{p}} = 1.24 \times 10^{-25}\, T_4^{-0.878}, \qquad \frac{4\pi j(\mathrm{H}\alpha)}{n_\mathrm{e} n_\mathrm{p}} = 2.87 \times 1.24 \times 10^{-25}\, T_4^{-0.938} \tag{4}$$

in erg cm^3 s^{-1} (`lines.f`:219–221; `src/EmissivityCalculator.cpp`:224–229), the Hβ coefficient attributed in both to Storey & Hummer (1995) and the 2.87 to Osterbrock's case-B table. Since the two exponents differ, the ratio drifts as $T_4^{-0.060}$ rather than staying at 2.87, which is what makes the fit work as well as it does.

A solved level population. CLOUDY has no emissivity table for hydrogen at all. Its hydrogen line data are reduced Einstein A coefficients for $n' \to n$ transitions (`data/hydro_tpn_n100.dat`, to $n = 100$ by default, with an $n = 2000$ set shipped alongside), and the line emission is whatever the solved H-like iso-sequence level populations give. The default structure is ten resolved levels (`iso.cpp`:192), raised by the `database h-like resolved/collapsed levels` command. Nothing is interpolated, so the low-density limit is a result of the solve rather than the edge of a grid, and the same applies at the top: CLOUDY computes through $n_\mathrm{e} = 10^{13}$ cm^{-3}, where a table-reading code has left its data behind. The price is that the answer is only as good as the level structure, which its own test suite states explicitly (below).

The three tabulation-derived families can be compared against one common standard, because MOCHII's parse of `e1bx.d` is the same Storey–Hummer case-B table that four of the six table- or fit-based codes ultimately cite. Table 4 evaluates each Hβ prescription against it.

Table 4: Each code's Hβ emissivity divided by the Storey & Hummer (1995) case-B table value at $n_\mathrm{e} = 100\,\mathrm{cm}^{-3}$, computed from the coefficients in the sources. IONIZE and CMACIONIZE share Eq. (4); TORUS and MOCASSIN's iteration path share $10^{-0.870\log_{10} T_\mathrm{e}+3.57}$; MOCASSIN's output path uses $2530\,T_\mathrm{e}^{-0.833}$ while the accurate fit sits commented out one line above it (`output_mod.f90`:2060–2063). All six temperatures listed are nodes of the table, so the three codes that interpolate it return the table value there and the interpolation scheme does not enter; what it costs between nodes is measured in Table 5.

T_e / K	MOCHII, MAPPINGS V 5.2.1, M^3	IONIZE, CMACIONIZE	TORUS, MOCASSIN (iteration)	MOCASSIN (output)
5 000	1.000	1.036	1.022	0.954
7 500	1.000	1.011	1.001	0.948
10 000	1.000	1.004	0.996	0.954
15 000	1.000	1.010	1.005	0.977
20 000	1.000	1.026	1.023	1.005
30 000	1.000	1.065	1.065	1.062

Two readings follow. The fits are good: over the range an H II region actually occupies, both fit families reproduce the table they are fitted to within a few percent, so the analytic route is a representational choice rather than an accuracy defect. And MOCASSIN's reported absolute Hβ is systematically 4.6–5.2% below the very table the same routine interpolates for its ratios, across the whole nebular range. That matters for this project specifically: MOCHII takes Hβ

from the table and MOCASSIN from a fit below it, so a few percentage points of any $L(\mathrm{H}\beta)$ comparison between the two codes is a coefficient artifact rather than physics. Line *ratios* are immune, because MOCASSIN divides by its own Hβ before writing them.

What the interpolation scheme costs. Reading a table exactly at its nodes is not the same as agreeing between them, and the three codes that interpolate the Storey–Hummer grid do it three different ways. MOCHII is bilinear in $(\log T_\mathrm{e}, \log n_\mathrm{e})$ on $\log j$, i.e. a power law within each cell (`sh95_mod.f90`:162–175). MAPPINGS V 5.2.1 and M^3 take a natural cubic spline in $\log_{10} T_\mathrm{e}$ of $\log_{10} j$ and interpolate linearly in $\log n_\mathrm{e}$ (`hydro.f`:255–265, 435–459). MOCASSIN's output path uses a five-point Lagrange polynomial in $\sqrt{T_\mathrm{e}}$ and $\log_{10} n_\mathrm{e}$ applied to $\log(j\sqrt{T_\mathrm{e}})$, dividing the smoothing factor out afterwards (`output_mod.f90`:2234–2366, `data maxv/4/ ni/2,3,4,5,6/`) — the scheme Storey & Hummer's own interpolation program uses.

Table 5 evaluates the three on the same case-B table. The spread is identically zero at the grid nodes and reaches a few tenths of a percent between them, largest where the grid is coarsest and at the high densities where the emissivity turns over. Two observations follow. The choice of scheme is not a material source of disagreement between these codes: at 0.1–0.3% it is an order of magnitude below the offset of MOCASSIN's fitted Hβ (Table 4) and below the 0.4–3.6% of the analytic fits. But the error of the low-order scheme is *one-sided*, which is the property that matters: the chord lies below the curve, so MOCHII's bilinear form is low at every off-node point rather than scattering about zero, and integrating over a model cannot average it away.

Because MOCHII's $L(\mathrm{H}\beta)$ sits below the published benchmark values, that bias is the natural suspect, and it can be measured on the converged models rather than estimated. Re-evaluating the same converged state — leaf by leaf, at each leaf's own $(T_\mathrm{e}, n_\mathrm{e})$ — with the two higher-order schemes in place of the bilinear one changes the volume-integrated $L(\mathrm{H}\beta)$ by

$$-0.061\% \ \ (\mathrm{HII40}), \qquad -0.126\% \ \ (\mathrm{HII20}), \tag{5}$$

identically against the spline and against the Lagrange form. The bias is indeed systematic and not statistical: of the 136 168 emitting leaves at HII40 *every one* is biased in the same direction, with a median of −0.065% and a worst case of −0.217%; at HII20, 93.5% are, with a median of −0.130%. HII20 carries twice the bias of HII40 because its temperature falls in the wider 5000–7500 K cell of the table where the chord error is largest. The Balmer decrement is less sensitive still, $\leq 0.05\%$, since the bias is common to the two lines and divides out of the ratio.

Set against a deficit of 3–5% (Section 11.2), the interpolation algorithm therefore accounts for one part in fifty at HII40 and one part in thirty at HII20. The remainder is the photon budget: the converged models return 0.950 and 0.970 of Q_H as hydrogen recombinations, with helium taking $0.112\,Q_\mathrm{H}$ at HII40, and $L(\mathrm{H}\beta)$ follows that budget rather than the emissivity table. Removing the one-sided bias — a spline or the Storey–Hummer $\sqrt{T_\mathrm{e}}$-smoothed form in place of the chord — is worth doing because a systematic error should not be left in a normalization quantity, but it will not move $L(\mathrm{H}\beta)$ perceptibly.

What a solved atom says about the tables. Having one code in the comparison that does not read the tables makes it possible to ask how good the tables are, which is otherwise unanswerable from inside this family. Two measurements were made with CLOUDY c25.00 for this memo, both single-zone, constant-temperature, hydrogen-only models with

Table 5: Spread among the three table-interpolation schemes: the largest of the three $j(\mathrm{H}\beta)$ values divided by the smallest, in percent, on the Storey & Hummer (1995) case-B table. Rows at 5 000, 10 000 and 20 000 K and the column at $n_e = 10^2\,\mathrm{cm}^{-3}$ are grid nodes, where all three schemes return the tabulated number and the spread vanishes by construction. 7024 and 8211 K are the converged benchmark temperatures.

	n_e / cm^{-3}			
T_e / K	10^2	3×10^3	10^5	3×10^6
5 000	0.000	0.043	0.000	0.279
7 024	0.114	0.105	0.092	0.205
8 211	0.100	0.096	0.082	0.172
10 000	0.000	0.024	0.000	0.145
13 693	0.042	0.040	0.047	0.112
20 000	0.000	0.006	0.000	0.074

30 resolved and 70 collapsed levels, run in the configuration of its own case-B reference test (`case b hummer no photoionization no pdest`, `no scattering escape`, `no induced processes`).

First, at the table's own grid point, $T_e = 10^4$ K and $n_e = 10^2\,\mathrm{cm}^{-3}$, the solved populations sit slightly *below* the tabulated case-B values, and by an amount that grows along the series:

	Hβ	Hα/Hβ	Paα/Hβ
Storey & Hummer (1995)	1.2350×10^{-25}	2.8640	0.3387
CLOUDY c25.00	1.2321×10^{-25}	2.8452	0.3319
difference	−0.23%	−0.65%	−2.00%

This is the quantitative version of the statement its test deck makes in words: the deviation is attributed there to the principal quantum number specification, 30/70 in the run against 100/1 in the table. The practical reading for a table-based code is reassuring in the abstract and cautionary in the detail — the normalization line agrees to a quarter of a percent, while a ratio between widely separated series does not, at the level a Balmer/Paschen decrement measurement would notice.

Second, and this is what settles the question the table cannot answer, the same model run down to $n_e = 10^0\,\mathrm{cm}^{-3}$ shows the hydrogen coefficients essentially flat, which is the measurement quoted in the density paragraph above. Its own validation suite tests $\log n_e = 2, 4, 5, 8, 10$ and 13; the lowest is the Storey–Hummer floor, because below it there is nothing tabulated to validate against. A code that solves the populations does not need the floor, and a code that reads the table needs a defensible rule at it.

Case A and case B. This is where the six differ most, and where the ordering is not the one the codes' ages would suggest. MAPPINGS V 5.2.1 has the most physical treatment of any code here: the case parameter is not a switch but a *continuous* quantity computed zone by zone from the Lyman-γ escape probability, $\mathrm{casab} = 1 - e^{-0.8597x}$ with x the line-transfer path multiplier (`functions.f`:21–71), and the code interpolates logarithmically between the case-A and case-B tables at that value. The same escape quantity that sets the parameter also attenuates the lines in the diffuse field, so the treatment is self-consistent. M^3 inherits the routine and the data but not the logic: its three-dimensional driver hardcodes `caseab(1)=1.0d0` with the escape-probability call commented out one line below (`calpdf.f`:45–46, 54–55), so the connected path is pure case B everywhere and the Lyman series above Lyα is identically zero even in optically thin gas. MOCHII sits between the two: two explicit parameters, `par%case_ab` for the ionizing continuum and `par%line_case` for the line tables, both defaulting to B, with the second added precisely because tying them together silently loaded case-A line tables whenever the diffuse field was switched on.

The other three each conflate the two questions in a different way. IONIZE has it right by convention and says nothing about it: its lines are case B (hardcoded, no switch) while its continuum uses the case-A total recombination coefficient with an explicitly transported diffuse field — the same pairing MOCHII arrived at, with no parameter and no comment marking the distinction. CMACIONIZE hardwires case-B line coefficients while the continuum case follows from two uncoordinated defaults, a diffuse handler that is off and a recombination coefficient that is the case-A total. CLOUDY is in a category of its own here, and it is the cleanest of the seven: the case is neither a table nor a flag on the data but a *transfer condition*. `case b` sets the minimum Lyman-line optical depth to 10^5 and case A or C to 10^{-5} (`parse_caseb.cpp`:19–41), and `rt_tau_reset.cpp`:137–144 then clobbers the Lyα optical depth to that value while every other line is transferred normally. The command also carries three options whose only purpose is to *emulate* the tabulated assumptions — `hummer` ("no collisions from n=1, 2"), `no photoionization` and `no pdest` — and its own case-B reference test uses all three, plus `no scattering escape` and `no induced processes`, with 30 resolved and 70 collapsed levels. The test's closing comment is the honest summary of what a case-B table is: "This is close to the best and most complete model of H I that the code can do. The predicted results still deviate from the tabulated 'Ca B' results because of the principal quantum number specifications (30/70 here; 100/1 in table)." In other words CLOUDY has to switch physics off to approach the tables, and the same test finds departure coefficients of 0.46–1.79 for levels $n = 30$ to 530 at $n_e = 10^2\,\mathrm{cm}^{-3}$, which is why the solved populations are the more general statement.

TORUS exposes a `caseB` flag that changes neither the lines nor the transport but replaces the hydrogen recombination coefficient with a temperature-independent constant $2.7 \times 10^{-13}\,\mathrm{cm}^3\,\mathrm{s}^{-1}$, and warns rather than refuses when it is combined with a transported diffuse field. MOCASSIN offers case B only.

Density dependence, and what happens below the table. MOCHII, MAPPINGS V 5.2.1, M^3 and MOCASSIN's output ratios carry it; IONIZE, CMACIONIZE, TORUS and MOCASSIN's iteration path do not; CLOUDY carries it inherently, since the populations are solved rather than read. That difference has a consequence the table-based codes have to answer for: the Storey–Hummer grids begin at $n_e = 10^2\,\mathrm{cm}^{-3}$ (the original CDS files as well as the repackaged ones), so every code that reads them needs a rule below that density, and MOCHII's rule is to clamp (`sh95_mod.f90`:163). Two measurements bound what the clamp costs. From the bottom slope of the table itself, extrapolating the collisional term linearly in n_e, the $n_e \to 0$ error is $\leq 0.08\%$ for every H I and He II line in it. Independently, CLOUDY — which has no floor — gives +0.065% for Hβ, +0.108% for Hα and +0.03 to +0.05% for the members up to $n = 17 \to 2$ between $n_e = 10^2$ and $10^0\,\mathrm{cm}^{-3}$ at 10^4 K. The two agree that for hydrogen the low-density limit is already reached at the table's first column, and the cost of the clamp on an integrated luminosity is smaller still: replacing it with the $n_e \to 0$ limit moves $L(\mathrm{H}\beta)$ by −0.0002% at HII40, −0.0003% at HII20 and −0.031% in the low-density circumgalactic demonstration where every emitting cell is below the floor.

Table 6: Hydrogen recombination lines, one row per code. "Case mechanism" is how the code decides between case A and case B, which is the axis on which no two of them agree.

Code	Emissivity source	Case mechanism	n_e depend.	H collis. term	Transported
MoCHII	SH95 case-A/B tables, bilinear in $(\log T_e, \log n_e)$	`line_case`, separate from the continuum `case_ab`	yes	optional, 25-level atom	no
MAPPINGS V 5.2.1	SH95 absolute Hβ + 90 ratios, spline in $\log T_e$	continuous, from the Lyγ escape probability	yes	yes, with case-dependent cascade	yes, outward-only with escape factors
M^3	the same routine and data	hardcoded case B in the 3D driver	yes	yes, as above	5–200 eV band only, so Lyman lines only
Mocassin	ratios frozen at (10^4 K, 10^2) in the iteration; SH95 interpolated at output	case B only, no switch	output ratios only	no	energy removed, photon not propagated
IONIZE	analytic fits to SH95, Eq. (4)	case B hardcoded, continuum case A	no	no	no
CMacIonize	the same coefficients	case B hardwired; continuum case from two uncoordinated defaults	no	no	no
Torus	ratios frozen at (10^4 K, 10^2) times a fitted Hβ	`caseB` changes only the recombination coefficient	no	no	yes, into the diffuse redraw spectrum
Cloudy	none: solved H-like level populations, $n \leq 100$ by default	a Lyman-line optical depth, 10^5 against 10^{-5}	inherent	inherent	1D line transfer

Helium is the exception, and it is not a small one. The Porter et al. (2012/2013) He I table starts one decade lower, $n_e = 10^1\,\mathrm{cm}^{-3}$, and its 10833 Å line does not flatten there: across the lowest decade of its own grid the coefficient moves +19.1%, against $\leq 0.8\%$ for the five singlet lines in the same table. Cloudy confirms the effect independently, +10.5% from $n_e = 10^1$ to $10^2\,\mathrm{cm}^{-3}$ and a further +1.1% from 10^0 to 10^1, against $\leq 0.04\%$ for 5876 Å. The reason is structural rather than numerical: the lower level of 10830 is the metastable 2^3S, whose population is set collisionally, so this line has no density-independent limit in the regime the table covers — which is precisely why the 2^3S population is computed explicitly elsewhere in MoCHII rather than inferred from a case-B emissivity. Clamping it is therefore a real error in diffuse gas, at the ten-percent level for $n_e < 10\,\mathrm{cm}^{-3}$, and the honest statement is that the tabulated 10833 emissivity should not be used below the Porter floor at all. The cost is small in an H II region and not small in a planetary nebula: from the same table, $j(\mathrm{H}\beta)$ rises 0.4% from $n_e = 10^2$ to $10^4\,\mathrm{cm}^{-3}$ at 10^4 K and 1.6% to 10^6, while $\mathrm{H}\alpha/\mathrm{H}\beta$ falls 2.0% by 10^6 and 5.7% by 10^8. This is the same axis as the metal-line one of Section 8.4, and two codes are internally inconsistent across it: CMACIONIZE and IONIZE both solve their metal lines from level populations at the local n_e while leaving the hydrogen lines density-independent.

Collisional excitation. MAPPINGS V 5.2.1 and M^3 are the only codes that add a collisional term inside the line calculation itself, and they do it carefully: the excitation rate to each level (Anderson et al. 2000 collision strengths for $n \le 5$, Giovanardi et al. fits for $6 \le n \le 16$) times case-dependent cascade branching fractions read from their own data file, added to the recombination brightness (`hydro.f`:756–812), with the same energy booked as cooling and not double counted. MOCHII has the equivalent as an option (`par%h_coll_effects`: a 25-level H I atom solved collisionally, its Balmer output written as separate rows so the recombination row stays a pure recombination quantity). CLOUDY needs no such term as an addition: collisional excitation is one of the rates in the level solve, which is what makes its Balmer emission depart from case B at all. IONIZE, CMACIONIZE, MOCASSIN and TORUS add nothing — consistent with the fact, recorded in the cooling audit, that none of them carries H I collisional-excitation cooling either.

Transport. The lines are diagnostics in MOCHII, IONIZE and CMACIONIZE: nothing re-emits a Balmer photon, and in CMACIONIZE the in-run emissivity call is commented out altogether, leaving a separate post-processing mode as the only route. MOCASSIN occupies a middle position that is easy to misread: the recombination-line energy is summed into the diffuse budget so that it correctly leaves the ionizing field, but a packet that lands in the line channel is given $\nu = 0$ and returns without being propagated — the energy is removed, the photon is not transported. TORUS genuinely transports them, writing the lines into the diffuse redraw spectrum so an absorbed ionizing packet can re-emerge as an Hα photon. MAPPINGS V 5.2.1 transports all of them into its outward-only diffuse field with escape and dust-retention factors. M^3 transports the subset that falls in its hardcoded 5–200 eV Monte Carlo band, which admits the Lyman series and excludes every Balmer and redder line, and it applies no escape factor to them.

What each code refuses to compute. IONIZE and CMACIONIZE return zero emissivity unless $x_{\mathrm{HI}} < 0.2$ and $T_e >$ 3000 K, so their emissivity maps are blank across the ionization front and through the neutral gas by construction; CMACIONIZE's Python twin uses 0.25 where the compiled code uses 0.2. MOCHII warns once when a cell exceeds the 30 000 K table edge and clamps rather than extrapolating. MAPPINGS V 5.2.1 zeroes all hydrogen lines outside $10 < T_e < 10^6$ K and extrapolates its ratio tables above 3×10^4 K, where they end.

Helium, for contrast. The same audit on helium separates the codes further. MOCHII carries Storey–Hummer He II tables in both cases and Porter et al. (2012/2013) He I emissivities; MAPPINGS V 5.2.1 and M^3 carry the same He II calculation and Porter et al. (2007), an earlier release from the same group; MOCASSIN uses a case-A He II table for the transported channel, which is the correct use of a case-A table for photons that ionize hydrogen. TORUS loads three helium tables at startup — Storey–Hummer He II, Benjamin et al. (1999) He I, and a Storey–Hummer H I table it never uses — and the routine that would consume them is commented out in its entirety along with both of its call sites, so TORUS emits no helium recombination line of any kind.

Two small findings, recorded for completeness. MOCASSIN adds its Lyman-α luminosity onto an existing slot of the H I array rather than a spare one, in both of its paths, and in the output path that slot reaches the user: the shipped HII40 result reports a row at $24.93\,L(\mathrm{H}\beta)$ carrying the $30 \to 8$ wavelength of 62 802 Å, which is Lyman α under a mid-infrared label. And in both MAPPINGS V 5.2.1 and M^3 the He II collisional add-on is evaluated with hydrogen's nuclear charge (`atom=2` followed by `nz=mapz(1)`, `hydro.f`:789–791), so its transition energy carries $Z^2 = 1$ instead of 4. The practical effect is negligible, since He$^+$ ground-state excitation is Boltzmann-suppressed at nebular temperatures, but it would matter in coronal gas.

10.3 Images and observing geometry

MOCHII writes leaf state and line-emissivity blocks to HDF5/FITS. Its Python reader deposits each AMR leaf with exact pixel-area overlap, so surface-brightness integration recovers the three-dimensional luminosity to machine precision. It also supports slab cuts and EM-, electron-, hydrogen-, or volume-weighted field maps.

For the transported FUV/EUV field, an extra converged-state pass applies observer peel-off. Stellar and diffuse emission contribute direct $e^{-\tau}$ terms, every dust interaction contributes an HG-weighted scattered term, and the optical-depth walk uses the same gas and dust opacity as the rate calculation. Multiple observers, unattenuated direct images, and bin-resolved cubes are available. The direct thin-medium gate matches the analytic image-integrated flux by +0.15% under the current defaults (exact hydrogenic cross sections, threshold-aligned bins).

M^3 outputs cell-resolved physical quantities and line emissivities, and its published applications construct spatial line maps from them. This is adequate for optically thin line projection. The public driver does not show a comparable general next-event estimator for arbitrary observers, nor the separation of direct and dust-scattered continua. Thus M^3 is richer in the number of intrinsic lines, while MOCHII is richer in the chain that turns a three-dimensional state into calibrated synthetic observations.

11. Convergence, verification, and maturity

11.1 Convergence criteria

M^3 updates the radiation field, solves each illuminated cell, and counts cells whose temperature and H$^+$ changes meet the selected tolerances. The run stops when a requested fraction of ionized cells has converged or after the iteration limit. Its public defaults use percent- level cell changes and increase the packet count when convergence stalls. This is intuitive, but the result is sensitive to how the ionized-cell mask and required fraction are chosen.

MOCHII reports both cell-max and volume-integrated pairs:

$$\delta_x^{(V)} = \frac{\sum V\,|\Delta x_{\mathrm{HII}}|}{\sum V x_{\mathrm{HII}}}, \qquad \delta_T^{(V)} = \frac{\sum V n_{\mathrm{e}}\,|\Delta T_{\mathrm{e}}|}{\sum V n_{\mathrm{e}} T_{\mathrm{e}}}. \tag{6}$$

The volume criterion suppresses front-cell jitter by the small volume of the front and excludes temperature oscillations in nearly electron-free cells. In the FUV dusty test, cell maxima remained at 0.1 and 6.4 while the volume measures reached stable Monte Carlo floors of 2.7×10^{-3} and 6.6×10^{-3}. Thirty further iterations changed the ionized volume by only 0.014%. This is a stronger convergence diagnostic for turbulent, clumpy grids, provided the tolerance is set just above the packet-noise floor.

11.2 Validation evidence

The MAPPINGS lineage has the stronger external maturity record. M^3 inherits decades of MAPPINGS development and has been published against the HII20, HII40, and PN150 Lexington/Meudon standards. Its three-dimensional blister, bipolar, and fractal applications demonstrate the importance of geometry for spatially resolved line diagnostics [1, 2]. MOCHII is newer, but its current repository contains unusually granular gates:

- analytic H/He attenuation and rate integrals;
- a Strömgren sphere against an exact one-dimensional reference;
- AMR versus uniform and solution-driven refinement versus native high resolution;
- H/He thermal balance against an independent Gauss–Seidel reference;
- direct CHIANTI fit evaluation and PyNeb line-ratio checks;
- MOCASSIN HII20/HII40 state and line comparisons;
- case-A/case-B diffuse-field bracketing;
- dust absorption/scattering brackets and IR energy closure;
- direct peel-off flux against an analytic observer solution; and
- exact three-dimensional emissivity-to-map luminosity conservation.

These gates provide strong internal evidence and make regressions easy to localize. They do not yet replace the independent user base and long publication history of MAPPINGS. The correct conclusion is that the MAPPINGS lineage is more mature externally, while MOCHII is more explicit and fine-grained in its current numerical audit trail. MAPPINGS V 5.2.1 does not have a Monte Carlo convergence problem: zone step control and the global shock/precursor iteration dominate instead. Its decades of use and broad model suite are a major maturity advantage, but the inspected 5.2.1 tree does not provide a gate hierarchy as explicit as the current MOCHII test directories. A controlled local benchmark is still required before attributing any line difference to geometry rather than to the 5.1.13-versus-5.2.1 atomic and solver divergence.

12. Software architecture and maintainability

MOCHII uses Fortran modules, allocatable leaf arrays, a single namelist, MPI-3 shared memory, HDF5/FITS sections, and a data-driven species registry. New atomic products are generated offline by Python tools and carry machine-readable provenance. A new ion normally requires no new solver code. The transport, ion balance, thermal balance, lines, dust, and output layers are separated modules. This architecture substantially reduces the cost and risk of adding physics.

The public M^3 tree retains the Fortran 77 MAPPINGS architecture: large common blocks, compile-time maxima, interactive input, and many global arrays. Several historical `montph7_*` variants duplicate large parts of the transport driver. This structure has the advantage of preserving a mature code base and its full menu of MAPPINGS models, but it makes a change to the 3D algorithm harder to propagate and test. The available Sphinx input/output documentation is sparse, so reliable use often requires reading the source and reproducing the authors' setup.

The architectural verdict is therefore clear: MOCHII is easier to extend, audit, automate, and connect to simulation data. M^3's value lies primarily in the depth of the inherited physical library, not in modern software structure.

MAPPINGS V 5.2.1 shares the same common-block and interactive heritage, but its organization is cleaner than the collection of duplicated M^3 `montph7_*` experiments and it has a single public `map52` model selector. Runtime preference files make the 16- or 30-element atom sets configurable without recompilation. Nevertheless, automated model grids require scripting interactive input and parsing many ASCII outputs; the current documentation overview explicitly remains incomplete. For maintainability the ordering is MOCHII first, MAPPINGS V 5.2.1 second, and the experimental M^3 3D fork third.

13. Suitability by science case

Table 7: Recommended code by present science requirement.

Science requirement	Preferred code	Reason
Clumpy or porous H II region from a TNG/RAMSES snapshot	MOCHII	AMR, front refinement, and simulation-oriented I/O.
Ionization-front position, shadows, and clump skins	MOCHII	Adaptive resolution and the analytic absorption estimator.
Dusty Strömgren structure and UV attenuation	MOCHII	Gas–dust competition and validated HG scattering in the 3D band.
PAH rings, IR SEDs, or 8–160-μm maps	MOCHII	SEDust, leaf dust emissivities, and exact absorbed-energy normalization.
Spatially resolved IFU line maps	Usually MOCHII	AMR emissivities and flux-conserving projection; verify that the required ions exist.
Very broad integrated optical/UV/IR line inventory	MAPPINGS V 5.2.1	Current full 30-element MAPPINGS preference set and deterministic detailed spectrum.

Table 7 continued.

Science requirement	Preferred code	Reason
Compact high-density nebula	MAPPINGS V 5.2.1 if 1D; M^3 if 3D	Both feed density-dependent multilevel cooling back into T_e; geometry decides.
Hard PN-like ionizing spectrum	MAPPINGS V 5.2.1 if 1D; M^3 if 3D	More high charge states; M^3 additionally has published PN150 3D validation.
Non-Maxwellian κ electron experiment	MAPPINGS V 5.2.1 or M^3	Both MAPPINGS-derived trees have native κ rate support.
One-dimensional radiation-pressure-confined nebula	MAPPINGS V 5.2.1	Isobaric photoionization can include the accumulated radiation pressure.
Finite source lifetime or source-off ionization history in 1D	MAPPINGS V 5.2.1	P6/P7 connect finite-age modes to the time-dependent ion solver.
Steady 1D radiative MHD shock and precursor	MAPPINGS V 5.2.1	S5 couples adaptive post-shock cooling to an iterated auto-preionization solution.
CIE, NEQ, or PIE cooling-curve grid	MAPPINGS V 5.2.1	Dedicated CC, NC, and PP drivers and a broad general-plasma ion inventory.
Controlled test of modern CHIANTI v11 coefficients	MOCHII	Versioned fitted data and direct fit-error gates.
Three-dimensional time-dependent ionization	None	Both public 3D drivers are equilibrium solvers; MAPPINGS V 5.2.1 time dependence is 1D.
Three-dimensional radiative shock	None	MAPPINGS V 5.2.1 has a strong 1D shock mode, but it is not coupled to either 3D transport architecture.
Molecular H_2/CO PDR	None	All three need a dedicated molecular chemistry and line-transfer layer.
Lyα or metal resonance-line profiles	None	Requires frequency-redistributing resonant transfer, e.g. LaRT.

14. Recommended development path for MoCHII

The comparison suggests a focused roadmap rather than wholesale replacement of either code.

1. **Put density dependence into the thermal cooling.** Reuse the generic n-level solver in the thermal loop for dominant coolants, or fit a suppression factor versus (T_e, n_e) around each critical density. This closes the most consequential gap relative to MAPPINGS.
2. **Extend the registry upward in ion stage.** Add the ions needed for PN and hard-field applications before adding marginal new low stages. The existing data-operation design is well suited to this.
3. **Add Na, Al, and Ni only when driven by a diagnostic need.** Their MAPPINGS implementations provide useful reference behavior, while the MOCHII pipeline should retain explicit provenance and ion-by-ion gates.
4. **Retain the present transport architecture.** The octree, analytic estimator, Cartesian DDA, threshold-aligned bins, and peel-off layer are major advantages and should not be replaced by the legacy M^3 packet synchronization model.
5. **Use MAPPINGS V 5.2.1 as the differential microphysics oracle.** Construct one-zone tests at fixed (T_e, n_e, J_ν) and compare ion fractions, heating/cooling terms, and individual line emissivities before comparing full 3D nebulae. Use M^3 only as the second-stage test of how that inherited microphysics behaves in arbitrary geometry.
6. **Perform a controlled common-physics benchmark.** Use the same uniform grid, source SED, abundances, dust-off setting, frequency range, and atomic subset. Compare J_ν, photo-rates, ion fractions, T_e, cooling budgets, and line emissivities separately. Only then compare total runtime and MPI scaling.

The most valuable hybrid would combine MAPPINGS V 5.2.1-like density-dependent microphysics with the current MOCHII geometry, transport, dust, and observer layers. This would preserve the strongest feature of each code without inheriting the principal numerical and software limitations of the other.

15. Final assessment

Capability	Stronger now	Confidence
Three-dimensional transport estimator	MOCHII	High
Adaptive spatial resolution	MOCHII	High
Dust scattering and IR/PAH observables	MOCHII	High
Observer imaging and data products	MOCHII	High
Atomic-data provenance and fit auditability	MOCHII	High
Element and ion-stage breadth	MAPPINGS V 5.2.1	High
Density-dependent thermal line cooling	MAPPINGS V 5.2.1/M^3	High
Hard PN/general plasma coverage	MAPPINGS V 5.2.1	High
Finite-age 1D photoionization	MAPPINGS V 5.2.1	High
Radiative MHD shock and precursor	MAPPINGS V 5.2.1	High
Noise-free 1D parameter-grid throughput	MAPPINGS V 5.2.1	High
External validation history	MAPPINGS lineage	High
Software extensibility and automation	MOCHII	High
Absolute line accuracy for every ion	Case dependent	Requires a common-data benchmark
Massively distributed memory ceiling	Architecture dependent	Requires a scaling experiment

For the principal intended application–dusty, structured, spatially resolved H II regions–MOCHII is the better current platform. For broad integrated emission-line diagnostics, dense nebulae, harder ionization conditions, finite-age 1D photoionization, and radiative shocks, MAPPINGS V 5.2.1 is the more complete physical reference. M^3 becomes preferable only when those MAPPINGS-like line and cooling strengths must be combined with arbitrary three-dimensional Cartesian geometry. None of these judgments is universal: first decide whether geometry, time history, shock dynamics, dust observables, density-dependent cooling, or ion-stage breadth is the controlling requirement.

16. Hard-spectrum (PN and AGN) extension plan

This section records the staged plan that follows from the comparison (kept as `docs/HARD_SPECTRUM_PLAN.md`, drafted 2026-07-18, not yet started). It uses the same staged-gate discipline as `docs/PLAN.md`.

16.1 Scope and the two tracks

"Hard spectrum" splits into two physically distinct regimes, kept as separate tracks because they need different physics:

- **PN track** — thermal central stars, $T_{\rm eff} \sim$ 75–200 kK. A 150 kK blackbody has kT = 12.9 eV; its photon flux at 300 eV sits $\sim$23 kT out on the Wien tail ($e^{-23} \sim 10^{-10}$ before the phase-space factors), and the lowest K-shell edge of an abundant metal (C, 290 eV) is never reached with significant flux. The MOCASSIN PN150 benchmark itself integrates its band only to `nuMax` = 24 Ryd = 326 eV. The PN track therefore needs *no inner-shell/Auger physics and no Compton terms* — it is band headroom, higher ion stages, the He II diffuse channels, and (at PN densities) density-dependent cooling.

- **AGN track** — a power-law/big-bump continuum extending to keV X-rays. Above $\sim$0.3 keV inner-shell absorption dominates the metal opacity and Auger multi-electron ejection couples non-adjacent ion stages; near $\sim$0.5–1 keV the Thomson cross section (6.65×10^{-25} cm^2) overtakes the steeply falling ($\sigma \propto E^{-3}$) photoabsorption of the residual neutrals, and Compton heating enters the energy budget. The first increment caps the band at $\sim$500 eV, where photoabsorption still dominates, and defers the Compton terms (Section 16.10).

Current baseline (v1.00-dev, 2026-07-18): band 13.6–100 eV (+FUV option); registry stages C 4 / N 3 / O 3 / Ne 3 / S 4 / Ar 4 / Mg 3 / Fe 4 / Si 5 / Cl 5 / Ca 5, with the compile-time stage ceiling `MAX_ST` = 6 in `species_mod` — the PN targets below fit it exactly, so no framework change is needed there; `te_max` = 5×10^4 K and the Tier-1 cooling fits are generated over 10^3–10^5 K; $\bar{g}_{\rm ff}$ is tabulated for $Z_{\rm ion}$ = 1–4 (`cooling_mod`, parameter `NGZ`); secondary ionization (`use_sec_ion`, Shull–van Steenberg 1985) is implemented but default off; the Owen-scrambled Sobol launch is available. The He II Storey–Hummer lines (case A and B), the He I Porter lines, and the diffuse He III ground channel already cover the He II 4686 diagnostic side, and the aligned-edge band builder (`ion_align_edges`, default on) with its graceful threshold-dropping logic is exactly the piece that makes many new in-band edges safe.

16.2 Reference assets (verified present)

Table 8: Local reference assets for the hard-spectrum work.

Asset	Use
MOCASSIN PN150 + PN75 benchmark inputs (`benchmarks/gas/{PN150,PN75}`)	primary 3D Monte Carlo gates; run MOCASSIN itself *converged* (the HII40 lesson). PN150 input read directly: 150 kK blackbody, $n_{\rm H}$ = 3000 cm^{-3}, $R_{\rm in} = 10^{17}$, $R_{\rm out} = 4.07 \times 10^{17}$ cm, band to 24 Ryd, abundances He 0.10, C 3×10^{-4}, N 10^{-4}, O 6×10^{-4}, Ne 1.5×10^{-4}, Mg 3×10^{-5}, Si 3×10^{-5}, S 1.5×10^{-5} (`abunPN150.in`) — note Mg and Si ARE in the composition, which drives their stage targets below.

Table 8 continued.

Asset	Use
Cloudy c23.01 test suite (`tsuite/auto/pn_paris*.in`, `nlr_paris.in`, `agn_lex00_u{0,1,m1}.in`)	1D cross-code gates: Meudon PN, AGN NLR, and the Lexington-2000 X-ray slabs. The `agn_lex00` inputs read directly: $n_{\rm H} = 10^5$ cm^{-3}, ϕ(H) = $10^{15.477}$ cm^{-2}s^{-1}, stop column 10^{16} cm^{-2}, and a *tabulated* SED (Cloudy `interpolate/continue` points) — so MOCHII reproduces the illumination exactly through the existing `spectrum_type` file route, without waiting for the AGN preset.
Cloudy AGN SED (`source/parse_agn.cpp`)	the standard AGN continuum parameterization (big-bump T, $\alpha_{\rm ox}$ anchored at 2500 Å–2 keV, $\alpha_{\rm uv}$, $\alpha_{\rm x}$) for a MOCHII preset.
Kaastra & Mewe (1993) Auger yields (`data/mewe_nelectron.dat`, `mewe_fluor.dat`)	electrons ejected per inner-shell vacancy, shell-resolved, machine-readable provenance for the Auger stage.
Verner & Yakovlev (1995) shell-resolved cross sections (`sigma_vy95` already in `photo_xsec.f90`)	inner-shell photoionization above the K/L edges.
CHIANTI v11.0.2 + Badnell 2023 RR/DR (in-tree)	levels/collision strengths and recombination for every new ion stage (coverage verified).

16.3 HS1: band headroom

Mostly validation; the code changes are one-line parameters.

1. `eion_max` (define.f90, default 100 eV) raised to 300–500 eV for hard runs. This matters even for stages the registry ALREADY tracks: the Ne IV→V threshold is 97.1 eV — just under the present cap — so today's band cannot photoionize to Ne V at all, and the new O/Ne thresholds (113.9, 126.2 eV) lie beyond it. Cross-section validity is not the issue (the Verner et al. fits extend to keV energies and H I/He II are exact hydrogenic); the work is verifying the pieces that ASSUME the band shape: the aligned-edge builder with $\sim$30–40 in-band thresholds once HS2 lands (recommend `nnu_ion` $\geq$ 64, 128 clean; the largest-remainder allocator and the metal-edge dropping logic already degrade gracefully), the FUV+ionizing two-segment bookkeeping, the diffuse `ion_bin_of` edge search, and the secondary-ionization band integrals whose $E_0 > 40$ eV branch becomes the COMMON case on a hard field.

2. $\bar{g}_{\rm ff}$: raise `NGZ` from 4 to 8 in `cooling_mod` (the van Hoof table already spans the required γ^2 range; the metal free-free sum then covers the new stages).

3. `te_max`: allow 10^5–10^6 K for X-ray-heated zones. Any ion used above 10^5 K needs its Tier-1 fit regenerated over the wider range (a flag in `chianti_cooling.py`, not new code); the bisection itself is range-agnostic.

4. Recommended hard-field defaults, documented in the example inputs: `use_sec_ion=.true.` (a 150 kK field makes $E_0 > 40$ eV photoelectrons routine, and X-ray fields more so) and `launch_sequence='sobol'` (the shielded, partially neutral zones where secondaries act are exactly where launch noise hurts most).

Gate HS1: the G0 pattern on the widened band — fixed-state analytic attenuation with a 150 kK blackbody, $\Gamma_{\rm HI/HeI/HeII}$ vs the analytic integrals bin by bin; $Q(>13.6)/Q(>24.6)/Q(>54.4)$ photon-rate ratios against exact Planck integrals to $< 0.1\%$ with aligned edges; and a spot check that the tabulated VFKY96 metal cross sections at 300–500 eV match the published fit values (guards against table-range assumptions in `photo_xsec`).

16.4 HS2: registry upward in ion stage

A data operation through the existing fitting pipeline:

Table 9: Target stages and the diagnostics they unlock.

El.	now→target	new thresholds [eV]	key new lines
C	4→5	C IV→V 64.5	C IV 1548/1550 (resonance — optically thin caveat)
N	3→5	47.4, 77.5	N IV] 1486, N V 1240
O	3→6	54.9, 77.4, 113.9	[O IV] 25.9 μm, O IV] 1400, O V] 1218
Ne	3→6	63.4, 97.1, 126.2	[Ne IV] 2422/2425, [Ne V] 3346/3426 + 14.3/24.3 μm, [Ne VI] 7.65 μm
S	4→6	47.2, 72.6	[S IV] 10.5 μm (new n-level file)
Ar	4→6	59.8, 75.0	[Ar V] 7.90/13.1 μm, [Ar VI] 4.53 μm
Mg	3→6	80.1, 109.3, 141.3	stage tracking for the PN150 composition (Mg 3×10^{-5}); [Mg IV] 4.49 μm, [Mg V] 5.61 μm as data permit
Si	5→6	Si V→VI 166.8	stage tracking for the PN150 composition (Si 3×10^{-5})

Fe/Cl/Ca hold (extend only on diagnostic demand; coronal [Fe VII+] deferred). All targets fit the existing `MAX_ST` = 6 ceiling exactly; `MAXLEV` = 40 / `MAXTR` = 600 and the 600-line `lines_mod` arrays already accommodate the new n-level models. Work items, all through `tools/fitting/`:

1. **Element files** (`make_element_data.py`): VFKY96 `PHOTO` rows exist in phfit2 for every target stage; Voronov CI covers all stages; Badnell 2023 RR+DR cover these isoelectronic sequences in the in-tree `clist_K` files (with the CHIANTI `RR2`/`DR2` fallback forms already supported where a fit is absent). **Charge exchange** above stage ∼4 is not in the Kingdon–Ferland fits: adopt the fast Dalgarno-type $\sim 1.92 \times 10^{-9} z$ cm^{3}s^{-1} recombination-only scaling used by Cloudy, with an explicit provenance note in each element file, or document neglect — decide once, apply uniformly (CX matters little in the hot interior where these stages live, but the decision must be recorded).

2. **Tier-1 cooling fits** (`chianti_cooling.py`) for each new stage, fit range matched to the chosen `te_max`.

3. **Tier-2 *n*-level files** for the diagnostic ions ([Ne IV], [Ne V], [O IV], [S IV], [Ar V], [Ar VI], N IV, N V, C IV); CHIANTI v11 carries level and collision data for all of them. Ions without CHIANTI level data are tracked without lines — the established Ar II pattern.

4. **PN abundance example** (`examples/`): the `abunPN150.in` values verbatim, so the PN150 gate input is reproducible from the repository alone.

Gate HS2: PyNeb ratio checks ion by ion at the established database-spread tolerances (sub-percent for well-determined optical pairs, up to ∼5–10% for the known compilation-spread classes): the [Ne V] 3426/24.3 μm temperature pair, the [Ne IV] 2422/2425 and [Ar IV] 4711/4740 density pairs, [O IV] 25.9 μm against its A-value limit; then an end-to-end hard-field smoke showing every new line in `_lines.txt` with physically sane ratios.

16.5 HS3: He II excited-channel diffuse photons

The PN He III zone is large, and case-B He II recombinations emit ionizing photons beyond the ground continuum already carried (`diffuse_mod` channel 3). The hydrogenic $n = 2$ cascade of He II produces three components, every one of which ionizes H I:

- He II Lyα at 40.8 eV — ionizes H I AND He I, the dominant piece;
- the He II Balmer continuum — its threshold is $Z^2\,\mathrm{Ry}/n^2 = 13.60$ eV, i.e. exactly at the H I edge; the aligned-grid `ion_bin_of` binary search (the 2026-07-18 bug fix) is precisely what makes this land in the correct bin;
- the He II two-photon continuum (0–40.8 eV), of which the > 13.6 eV part ionizes — sampled like the He I two-photon branch with its ionizing probability precomputed.

Implementation is the `hei_diffuse` pattern verbatim: a fifth channel in `dif_ch` weighted by the case-B He III recombination rate in each leaf, branch fractions from the hydrogenic 2s/2p cascade shares (Osterbrock & Ferland tables), energies fixed (Lyα), threshold-plus-exponential (Balmer continuum), or flat-sampled (two-photon). The quasi-random diffuse launch needs NO new dimensions — the channel pick d7 simply spans five values. Regime note: at HII40 the analogous He I channel moved T_e by only ∼1%, but there the He III zone was a small core; in a 150 kK PN the He III region fills much of the nebula and this becomes a first-order energy-budget term.
Gate HS3: channel energy conservation (emitted diffuse luminosity equals the case-B $n \geq 2$ recombination power of the He III zone); the PN150 He structure and T_e recorded with the channel off/on, attributed as one lever.

16.6 HS4: PN validation gates

MOCASSIN PN150/PN75, with the full-quality protocol that resolved HII40: MOCASSIN re-run CONVERGED (its stored reduced settings — 10^6 photons, 10 iterations, `convLimit` 0.05, 13^3 — are a moving target exactly as the HII20/40 baselines were), and MOCHII run with `conv_crit='vol'`, `nnu_ion` 64–128 aligned, $\geq 4 \times 10^7$ packets, `launch_sequence='sobol'`. The PN150 setup is fully specified by the benchmark input quoted in Table 8 (150 kK, $n_H = 3000$ cm^{-3}, shell 10^{17}–4.07×10^{17} cm). Compare, in order of diagnostic power:

1. T_e(H II) and the radial T_e profile (PN Te ~12–18 kK — also exercises the hotter thermal range);
2. the He I/II/III structure — the He II-edge (54.4 eV) photon budget is the PN analog of the He edge discretization that dominated HII40, so the aligned grid must show its value here;
3. the new high-stage fractions O III/IV/V, Ne III/IV/V, N III/IV;
4. the line table: L(Hβ), He II 4686/Hβ, [O III] 5007/4363, [Ne V] 3426, [Ne III] 15.5 μm, [O IV] 25.9 μm, [N II] 6584, [S IV] 10.5 μm, [Ar V].

Cloudy `pn_paris` as the 1D cross-check with identical SED, abundances, density, and dust off: zone-by-zone T_e and ion fractions along the radius plus ~20 integrated ratios. Because Cloudy solves density-dependent level populations in its thermal loop, the PN150 residual BEFORE stage HS7 directly measures how much the low-density Tier-1 fits miss at $n_H = 3000$ cm^{-3} — that number is the input to HS7, which is why HS4 deliberately precedes it.
Acceptance: land inside (or bracket) the published Lexington PN150 cross-code spread, with each residual attributed one lever at a time (binning, diffuse channels, density-dependent cooling, data vintage) exactly as the HII40 campaign did.

16.7 HS5: AGN SED input

An `ion_spectrum='agn'` preset implementing the Cloudy continuum (`parse_agn.cpp`; Mathews–Ferland form):

$$f_\nu = \nu^{\alpha_{uv}} e^{-h\nu/kT_{bump}} e^{-kT_{IR}/h\nu} + a\nu^{\alpha_x}, \tag{7}$$

with the X-ray normalization a fixed by α_{ox} through the 2500 Å–2 keV energy ratio (the 403.3 factor in the Cloudy source) and the standard defaults $\alpha_{uv} = -0.5$, $\alpha_x = -1$, IR cutoff at $kT_{IR} = 0.01$ Ryd. It enters `ion_band_mod` exactly like the Planck function — a spectral density integrated on the 32-point sub-grid in each band bin — and is normalized by the existing derive-or-rescale luminosity rules. The tabulated-file route (`spectrum_type='per_ev'` etc.) remains the reference path and already reproduces the `agn_lex00` gate SEDs verbatim (Table 8); the preset adds convenience and reproducibility for parameter studies. Also add an ionization-parameter helper to the rank-0 log, $U = Q_{ion}/(4\pi r^2 n_H c)$ evaluated at a reference radius, so slab setups map onto Cloudy inputs without hand conversion.
Gate HS5: the bin-integrated preset spectrum against Cloudy `save continuum` output for identical parameters (shape agreement to ~1% per bin); Q-ratio integrals exact; the derive/rescale bookkeeping re-verified on the preset (the ISRF-preset test pattern).

16.8 HS6: inner-shell absorption and Auger (AGN track)

The one genuine framework extension of the plan, in four parts:

1. **Shell-resolved opacity.** Above each K/L edge the total cross section becomes the VFKY96 outer-shell fit plus the Verner & Yakovlev (1995) inner-shell partials, $\sigma_{tot}(E) = \sigma_{outer}(E) + \sum_s \sigma_s(E)$. The `sigma_vy95` evaluator and the `PHOTO2` element-file rows already exist (the Cl path); the extension is carrying several shells with their edges in the aligned-band threshold list.

2. **Auger yields.** Kaastra & Mewe (1993) electrons per vacancy, shell-resolved, read from the Cloudy data file format (`mewe_nelectron.dat`: element, ion, shell, yield distribution): an inner-shell ionization of stage i lands at stage $i+n_\mathrm{e}$(shell) with the tabulated probability distribution over n_e. Fluorescence (Kα photon emission instead of one Auger electron, `mewe_fluor.dat`) is a small branch for low-Z elements; the first increment deposits it as local heat with a documented note rather than transporting the line photon.

3. **Cascade generalization.** `species_fractions` currently solves the strict adjacent-stage chain as a product form. Auger couples stage i to $i+2, i+3, \ldots$, so each element's stationary balance becomes a small $n_\mathrm{stage} \times n_\mathrm{stage}$ matrix problem ($M\mathbf{x} = 0$ with $\sum_j x_j = 1$); the partial-pivot elimination already used by `nlevel_mod` is reused directly. The chain path is KEPT as the default and the matrix path activates only when an inner-shell channel is present — the off-path stays arithmetically identical, the house rule for every extension.

4. **Energetics.** The Auger electron carries $E_0 = h\nu - E_\mathrm{th,shell} - \sum E_\mathrm{secondary\ vacancies}$ and is fast; it enters the EXISTING Shull–van Steenberg partition through the absorber-resolved excess-energy integrals with no new heating physics. Compton remains excluded at $\leq$500 eV.

Gate HS6: the X-ray-illuminated slab against the Cloudy `agn_lex00_u{m1,0,1}` set (three ionization parameters at $n_\mathrm{H} = 10^5$ cm^{-3}, tabulated SED, stop column 10^{16} cm^{-2}): ion-fraction profiles of C/N/O/Ne through the partially ionized zone, the T_e profile, and the H/He recombination lines, within the published Lexington-2000 cross-code spread. A secondary check: with Auger artificially disabled the slab must reproduce the HS2-era solution (isolates the new coupling).

16.9 HS7: density-dependent metal cooling in the thermal loop

Required at PN densities. The coolants that dominate at nebular temperatures quench in a specific order because their critical densities span five decades (approximate values at 10^4 K): [C II] 158 μm $n_\mathrm{crit,e} \sim 50$, [N II] 122 μm $\sim 3 \times 10^2$, [O III] 88 μm $\sim 5 \times 10^2$, [O III] 52 μm $\sim 3.5 \times 10^3$, [O II] 3729 $\sim 3 \times 10^3$, [S II] 6716 $\sim 2 \times 10^3$, [N II] 6584 $\sim 9 \times 10^4$, [O III] 5007 $\sim 7 \times 10^5$. At the PN150 density ($n_\mathrm{H} = 3000$, n_e up to $\sim$7000 cm^{-3} in the He III zone) the IR fine-structure lines are already partly quenched while the optical forbidden lines are not — the low-density Tier-1 fits therefore OVERESTIMATE cooling and bias T_e low, with the error growing toward compact PN and NLR densities.
Primary design, keeping "fitted offline, evaluated online":

1. `chianti_cooling.py` already solves the full n-level system to build its fits; extend it to emit $\Lambda(T, n_\mathrm{e})$ tables on a log-n_e grid ($\sim$7 nodes, 10^1–10^7 cm^{-3}) $\times$ the existing log-T range, one block appended to each `cooling_tier1_*.txt` behind a format-version flag (old files keep working).

2. `species_mod::metal_cooling` evaluates bilinear-in-($\log T, \log n_\mathrm{e}$); the memory cost is trivial ($\sim$60$\times$7 values for each ion) and the evaluation stays a handful of multiply-adds inside the bisection — the `species_ne` cache pattern applies unchanged.

3. The lowest-density node MUST reproduce the current fits to fit accuracy (regression guard), so HII20/HII40 and every recorded gate move by $< 0.1\%$.

4. Verification: an in-loop n-level solve for the dominant coolants on a small model, confirming table interpolation error $< 1\%$ in Λ over the used (T, n_e) range; the H-impact PDR cooling terms (`metal_cooling_H`) stay as they are (their two-level low-density form is correct in the PDR regime they serve).

Gate HS7: a two-zone analytic check (one low- and one high-density zone against the exact n-level cooling); PN150 T_e with the tables off/on (the residual measured at HS4 must shrink accordingly); HII20/HII40 unchanged to $< 0.1\%$.

16.10 Deferred items

Recorded once so they are conscious exclusions, not oversights: Compton heating/cooling and Thomson-scattering transport ($\gtrsim$0.5 keV; Cloudy is the reference implementation); Fe K fluorescence photon transport (the yield enters HS6 as local heat); coronal ions beyond ∼stage 8 ([Fe VII] 6087, [Fe X] 6374 need Fe stages 8–11 and a hotter thermal range); X-ray and UV resonance-line transfer (LaRT territory); grain destruction/sputtering in hard radiation fields (the PAH survival switches are a partial hook, not a destruction model); and time dependence / finite source age, where MAPPINGS remains the 1D reference.

16.11 Suggested order

Order	Stage	Framework risk	Blocking
1	HS1 band headroom	low	gates HS2+
2	HS2 registry upward	none (data)	gates HS4
3	HS3 He II diffuse channels	low	sharpens HS4
4	HS4 PN gates (MOCASSIN + Cloudy)	none	PN track done
5	HS7 density-dependent cooling	medium	PN accuracy
6	HS5 AGN SED preset	low	gates HS6
7	HS6 inner-shell/Auger	high (cascade)	AGN track done

HS7 can run in parallel with HS2–HS4 (independent code paths); the first PN150 pass deliberately precedes HS7, so it measures how much of the PN residual the density-dependent cooling must explain — the same one-lever-at-a-time attribution that worked for HII40.

17. Feasibility: time dependence, cooling flows, and shocks in MOCHII

Section 17. addresses the three capabilities where the public M^3 3D driver and MOCHII both trail MAPPINGS V 5.2.1: the connected time-dependent ionization history of P6/P7, the 1D cooling flow, and the S5 steady radiative shock with an iterated precursor (§3.2). The question is not whether these are useful — they plainly are — but how much each would cost to add to MOCHII, given its architecture. The three answers differ sharply: time dependence is a bounded project that the design accommodates naturally, a cooling flow rides on top of it, and a radiative shock is a separate one-dimensional radiation-hydrodynamic calculation that the Monte Carlo transport engine does little to help.

17.1 The architectural pivot

MOCHII's iteration loop transports Monte Carlo packets, tallies the cell radiation field, solves the *equilibrium* ionization and thermal balance in each cell (`solve_ion_cell`: a damped fixed point on n_e with a bisection on T_e), refreshes the opacity, and repeats until converged. The state solver is therefore a *root finder*: given the rates it returns the steady state. It integrates no $\mathrm{d}n_i/\mathrm{d}t$, carries no velocity field, and solves no fluid equations. That single fact sets the cost of each addition: one reuses the root finder almost unchanged, one adds a time integrator beside it, and one needs what the code does not have at all.

17.2 Time-dependent ionization: moderate, and the natural fit

Almost everything a finite-lifetime calculation needs already exists. The transport, the photoionization, recombination, collisional-ionization, and charge-exchange rate coefficients (already evaluated cell by cell), the tally, the octree, and the diffuse field are all reusable without change. The cell solver is already "given the rates, find the state"; the new piece is a sibling routine, "given the rates, the old state, and Δt, find the new state."

What is genuinely new is threefold. First, a stiff ordinary-differential integrator for the ionic populations and, where the thermal time is not short compared with the ionization time, the electron temperature — a backward-Euler or semi-implicit step in each cell, of the kind MAPPINGS V 5.2.1's `timion` performs, rather than the present root find. Second, an outer time loop with a step controller keyed to the atomic, cooling, and photon-absorption times. Third, a source light curve or a finite on/off lifetime as input.

The design accommodates this cleanly through operator splitting. MOCHII already performs one transport pass for each iteration, so the natural structure is an outer radiation step that holds J_ν fixed while the chemistry is substepped in each cell, with the radiation field recomputed once the ionization has advanced enough to change the opacity. An ionization front that advances while a source brightens, or recedes after it switches off, is a reachable target. The hard part is the stiff integrator and the step control, not the infrastructure.

17.3 Cooling flows: depends on which cooling flow

The term carries two meanings, and they price very differently.

In the MAPPINGS V 5.2.1 sense — a parcel of gas that cools radiatively as it moves down a prescribed density and velocity history — a cooling flow is a special case of time-dependent cooling on a fixed trajectory: switch off (or freeze) the photoionizing field, and integrate the non-equilibrium ionization and the radiative losses down a temperature ramp. Once the time-dependent solver of the previous subsection exists, this is a modest addition, because the cooling terms it needs — recombination, free–free, and the metal-line coolants through the Tier-1 fits and the Tier-2 n-level solve — are already present. Its one real prerequisite is the density-dependent cooling flagged as HS7 and in §8.4: a cooling flow crosses coolant critical densities, so a $\Lambda(T_\mathrm{e}, n_\mathrm{e})$ that suppresses forbidden lines correctly is required for it to be quantitative.

In the cluster or galaxy sense — a self-consistent inflow driven by its own radiative losses, with gravity and hydrodynamics — a cooling flow is a fluid-dynamics problem and lies outside the present scope without a hydrodynamic solver. The comparison with MAPPINGS V 5.2.1 concerns the first meaning.

17.4 Radiative shocks: hard, and off the architecture

The MAPPINGS V 5.2.1 S5 shock solves the Rankine–Hugoniot jump conditions and then integrates the post-shock cooling zone as a one-dimensional flow — mass, momentum, and energy conservation along the flow coordinate with radiative losses — while iterating an upstream photoionized precursor computed from the shock's own radiation to self-consistency. Its essence is a one-dimensional radiation-hydrodynamic flow, not a three-dimensional transport problem.

This is the poor fit. MOCHII has no velocity field, no fluid equations, and no flow coordinate. The post-shock structure is gas that moves and compresses; the static octree grid and the Monte Carlo photon transport represent none of that, and three-dimensional Monte Carlo transport buys almost nothing for a one-dimensional steady shock. The honest route is a separate one-dimensional radiative-shock module, a companion solver in the spirit of the 1D references MOCHII already uses for its gates: it would reuse the atomic and cooling data but not the octree or the Monte Carlo engine, and it would carry the jump conditions, an adaptive cooling-zone integrator, and the precursor iteration itself. That is real work, largely decoupled from the transport code, and — this is the key qualification — it would be a 1D shock, not a 3D radiative shock. A genuine three-dimensional radiative shock needs actual radiation hydrodynamics, which is a different code, not an extension of this one.

17.5 Summary and suggested order

Capability	Difficulty	Why
Time-dependent ionization	Moderate	Reuses all transport and rates; new work is a stiff cell ODE, a time loop, and a light curve. Operator splitting maps onto the existing one-transport-for-each-iteration loop.
Cooling flow (MAPPINGS V 5.2.1 sense)	Moderate	A prescribed-trajectory special case of the time-dependent solver; needs the density-dependent $\Lambda(T_e, n_e)$ of HS7.
Cooling flow (cluster sense)	Out of scope	Requires gravity and hydrodynamics.
Radiative shock (S5)	Hard	A separate 1D radiation-hydrodynamic module; the octree and Monte Carlo engine do not apply. Not a 3D shock.

A sensible order is: (1) time-dependent ionization first, as the most natural addition and the foundation for the rest; (2) the density-dependent cooling of HS7, needed for both PN accuracy and cooling flows; (3) the MAPPINGS V 5.2.1-sense cooling flow as a thin layer on top of (1) and (2); and (4) a decoupled 1D radiative-shock module last, and only if a shock diagnostic is actually required. The first three share the same time-integration and cooling infrastructure and reinforce the PN and hard-spectrum work of §16.; the fourth stands apart.

18. Implementation evidence consulted

The principal local sources used for this audit were:

- MOCHII: `docs/PLAN.md`, `docs/MoCHII_physics.tex`, `docs/MoCHII_UserGuide.tex`, the transport/grid modules under `src/`, and the quantitative gates under `tests/`.
- M^3: `README.md`, `lab/data/ATDAT.txt`, `src/mastercode/montph7.f`, `montph7_dust.f`, `crosssections.f`, `mcteequi.f`, `teequi2.f`, `equion.f`, `cool.f`, `multilevel.f`, `grainpar.f`, `hgrains.f`, `hpahs.f`, `dusttemp.f`, `timion.f`, and `shock2--shock5.f`.
- MAPPINGS V 5.2.1: `README.md`, `src/mappings.f`, `photo6.f`, `photo7.f`, `teequi.f`, `teequi2.f`, `timion.f`, `timtqui.f`, `cool.f`, `dusttemp.f`, `shock5.f`, `neqc.f`, `coolc.f`, `transferline.f`, `lab/mapStd.prefs`, `lab/mapFull.prefs`, and `docs/DocumentionOverview.html`.

The public M^3 source contains several historical or experimental driver variants. Statements in this memo refer to the main `mastercode/montph7.f` and explicitly selected `montph7_dust.f` paths unless otherwise stated. Statements about MAPPINGS V 5.2.1 refer to the connected public 5.2.1 `map52` P6/P7/S5/CC/NC/PP paths, not merely to routines present in the directory.

References

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